Uncertain Climate Policy as a Source of Macro-Financial Shocks: Evidence from Carbon Futures Volatility*

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Abstract

We study climate policy uncertainty through a macro-finance lens, focusing on the transmission of climate uncertainty shocks to aggregate and sectoral dynamics. Using monthly U.S. data from 2014 to 2025, we estimate a Bayesian Proxy-SVAR identified with an external instrument derived from the realized volatility of the ICE U.S. Carbon Futures Index, a market directly exposed to climate regulation. Unlike widely used textual and news-based indicators, this market-based proxy captures real-time reassessments of climate-policy risk priced by financial markets. We find that climate uncertainty shocks tighten financial conditions, raise market volatility and financial stress, and reduce output and stock market valuations, triggering an accommodative monetary policy response. Prices fall in the short run and rise above their pre-shock level over the medium run. At the sectoral level, we document heterogeneity along two complementary dimensions. A broad green–brown decomposition shows that transition risk-exposed (brown) sectors experience deeper and more persistent contractions, whereas less exposed (green) sectors recover faster. We perform a full sector-by-sector disaggregation of industrial production and valuation responses and match each sector to its EPA supply-chain emission intensity. This granular mapping reveals a systematic relationship between carbon intensity and the severity of climate uncertainty-induced contractions: higher-emission industries exhibit larger real and financial declines. Overall, climate-policy uncertainty priced in carbon futures markets acts as a financial amplifier, propagating primarily through volatility and valuation channels and reinforcing the asymmetric dynamics of the low-carbon transition.

Keywords: Climate Policy Uncertainty, Financial Transmission Mechanism, Bayesian Proxy-SVAR, Green vs Brown Sectors, Realized Volatility.

JEL Codes: E32, E44, Q54, G12.

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1 Introduction

The accelerating effects of climate change pose mounting risks to global macroeconomic and financial stability by altering production technologies, asset valuations, and the distribution of future economic outcomes (see Battiston et al. (2021)). Governments around the world have adopted increasingly ambitious environmental targets, yet the future path and credibility of climate policy remain highly uncertain (see Rogelj et al. (2023)). In the United States, this uncertainty has intensified during recent political transitions, most notably around the beginning and the end of the two non-consecutive Trump's presidential mandates, which have marked a sharp reversal in federal U.S. climate policy (see Besher et al. (2025)). Ambiguity over the timing, scope, and persistence of climate policy has created what is now referred to as climate (policy) uncertainty—a type of risk that can delay investment, raise financing costs for both private firms and the public sector, and ultimately amplify macroeconomic instability and financial stress following climate-related shocks (Barnett et al., 2020).

While a growing empirical literature documents the macroeconomic and financial effects of climate policy uncertainty, existing measures remain largely textual or news-based, relying on newspapers, media coverage, and policy-related narratives. These indicators provide valuable information about the salience and evolution of climate-policy debates and have substantially advanced our understanding of the role of climate-policy uncertainty. At the same time, the distinction between first-moment shocks, associated with revisions in expected climate-policy actions, and second-moment shocks, associated with time variation in the uncertainty surrounding those expectations, has received comparatively less emphasis in the literature. This distinction matters because first- and second-moment disturbances tend to occur together and are difficult to separate empirically, as shown by Cascaldi-Garcia and Galvao (2021) for news and uncertainty shocks. Moreover, relatively little attention has been devoted to market-based measures of climate-policy uncertainty and to the role of financial markets in assessing and transmitting climate-related risks. As a result, our knowledge of how climate-policy uncertainty is incorporated into asset prices, financial conditions, and ultimately macroeconomic outcomes remains limited. In this paper, we complement the existing textual approach by constructing a market-based measure of climate policy uncertainty derived from carbon futures markets and embedding it in a structural macro-finance framework. This approach allows us to trace how unexpected revisions in climate-policy risk priced by financial markets propagate through financial conditions, macroeconomic aggregates, and the cross-section of industries.

We make three contributions. First, we construct a market-based measure of climate policy uncertainty for the United States and validate it with a daily event study. Second, we show that shocks to this measure are contractionary and deflationary in the short run, while the price level rises above its pre-shock level over the medium run. Third, we link sectoral real and financial responses to carbon intensity: matching each industry to its EPA emission intensity and finding that higher-emission industries tend to suffer larger declines in output and valuations.

Given the latent nature of uncertainty about future climate-policy developments and their economic consequences, measuring climate transition uncertainty remains a major empirical challenge. Among the most widely used approaches, Gavriilidis (2021) and Känzig et al. (2025) construct climate uncertainty indices from major U.S. newspapers, building on the framework of Baker et al. (2016). Their measures track fluctuations in the intensity of the climate-policy debate by identifying

articles that jointly reference uncertainty, climate-related issues, and government policy.

Although the economic transition towards net-zero emissions is expected to entail short-term adjustment costs, unpredictable or inconsistent policy signals may generate even greater losses by undermining confidence and distorting the allocation of resources. A growing literature shows that climate *policy* uncertainty has measurable macroeconomic and financial effects. In particular, Känzig et al. (2025) demonstrates that climate uncertainty shocks have recessionary impacts, acting as supply-side disturbances that reduce output and investment through higher production costs and delayed adjustment in the real economy. Similar evidence is found in Palikhe et al. (2024), who develop an index of environmental policy uncertainty and use a VAR framework to show that higher uncertainty lowers firm-level investment in industries sensitive to environmental policies. Likewise, Noailly et al. (2022) construct a broader environmental policy uncertainty index using machine learning techniques and find that higher environmental policy uncertainty reduces low-carbon investment and venture-capital funding.

Based on a similar textual approach, Basaglia et al. (2025) analyze the firm-level implications of climate policy uncertainty. They document that heightened climate uncertainty leads to lower capital expenditures, innovation, and R&D spending—especially among carbon-intensive (brown) firms—and increases stock-market volatility. These results are consistent with the theoretical argument that predictable and credible climate policies can reduce transition costs and support long-term growth (Fried et al., 2021). Besides these macroeconomic and microeconomic perspectives, a growing literature has emphasized the financial transmission mechanisms for climate and transition risks, showing how they affect the pricing of financial assets, the cost of capital of firms, and portfolio allocation across sectors through the volatility and discount rate channels (e.g., Bolton and Kacperczyk, 2021; Pástor et al., 2022; Ardia et al., 2023). Complementing this evidence, Annicchiarico et al. (2026) show that financial frictions shape the transmission of business cycle shocks under alternative carbon-pricing regimes.

More generally, textual and market-based indicators provide complementary perspectives on climate-policy uncertainty. While textual and narrative sources (such as newspaper articles) offer valuable information on the intensity and evolution of climate-policy debates, market-based measures capture how climate-policy risk is assessed, priced, and transmitted by financial market participants through asset prices and financial conditions. As such, they can provide additional insights into the transmission of climate-policy uncertainty through financial markets and the broader economy.

For instance, consider a situation in which credible information about a potential tightening of climate regulation becomes available to market participants, but the probability of enactment remains unclear. In such cases, financial markets may immediately re-price assets exposed to transition risk, leading to sharp adjustments in derivatives-implied volatility or risk premia, even in the absence of contemporaneous price movements. Textual and market-based indicators may therefore respond differently to the same episode, reflecting their distinct informational content and measurement objectives.[1] In fact, our work is closely related in spirit to recent studies on

[1]Event-study evidence indicates that climate-policy news can be priced very quickly in traded assets. Using event studies on sectoral ETF data, Antoniuk and Leirvik (2024) find broadly positive cumulative abnormal returns around the adoption of the Paris Agreement, while the November 2016 U.S. presidential election is associated with a sharp negative repricing, with post-event cumulative returns below -2.5% across all sectors. By contrast, consistent with measurement differences across textual constructions, Gavriilidis (2021) reports that CPU reacted over all of early 2016 to the Agreement, which is consistent with the fact that their CPU index correlates only 0.41 with the

market-implied measures of climate uncertainty, such as the Carbon VIX developed for the EU Emissions Trading System, which captures carbon-price volatility (Fuchs et al., 2026).

We contribute to the literature by providing a market-based measure of climate transition uncertainty and by analyzing its macro-financial and sectoral effects on the U.S. economy. We construct an external instrument derived from the realized volatility of the ICE (Intercontinental Exchange) U.S. Carbon Futures Index—a financial market directly exposed to developments in emission-allowance regulation. This index tracks the futures prices of carbon allowance contracts and represents a market-implied indicator of expected carbon prices and, by extension, of climate policy stringency and volatility.

To isolate the unpredictable component of volatility, we remove its autoregressive structure and retain the innovation term, which captures unanticipated shifts in investors' perceptions of climate-policy risk. The resulting realized volatility innovation captures real-time revisions in policy expectations, offering a clean, high-frequency proxy for exogenous climate uncertainty shocks embedded in financial markets. We use this instrument to identify a monthly Bayesian Proxy-SVAR over the period 2014–2025, in which the VIX, the implied volatility index, serves as the financial variable through which climate transition uncertainty shocks are transmitted. This framework allows us to examine how climate-policy uncertainty interacts with broader market volatility and affects real activity and asset prices through financial markets. To assess whether our measure captures climate-policy uncertainty, we conduct a daily event study covering 167 climate-policy events. On event days, carbon-futures volatility rises to roughly twice its pre-event level, supporting the measure's climate-policy content.

In addition to the aggregate analysis, we add a sectoral dimension that has not been explored in the prior literature. We study heterogeneity in the transmission of climate uncertainty shocks along two complementary layers of disaggregation. First, we construct a broad green–brown decomposition to document differences between transition-risk-exposed and transition-resilient industries. Second, we move to a full sector-by-sector analysis of both industrial production (using the the complete set of NAICS-based industry groups in the Federal Reserve G.17 IP series) and sectoral valuation dynamics (using the Fama–French 49 Industry Portfolios). A key advantage of this granular structure is that it permits a systematic mapping between each sector and its EPA supply-chain emission intensity. Through this linkage, we can directly assess whether the severity of the sectoral responses to climate transition uncertainty shocks varies with the underlying carbon intensity of economic activity. This is an aspect that, to our knowledge, has not been quantitatively examined in previous studies of climate-policy uncertainty.

An unexpected increase in climate-policy uncertainty raises aggregate equity implied volatility and tightens financial conditions. Industrial production and stock market valuations decline, while monetary policy responds with an easing. The goods price level initially falls but rises above its pre-shock level over the medium run. The decline in the price–dividend ratio is particularly persistent, suggesting that the effects on asset valuations outlast the contraction in real activity.

Our findings are robust to alternative model specifications, including different lag structures, prior settings, and the selection of alternative macro-financial aggregates, and hold under both Local Projections (LP) and Bayesian Local Projections (BLP) frameworks. They also persist when the instrument is residualized on measures of carbon prices, energy-market conditions, policy

Wall Street Journal climate-change news index of Engle et al. (2020).

uncertainty and financial-market conditions. We also assess their sensitivity to the COVID-19 and Ukraine episodes.

When we extend our analysis to a sectoral dimension by investigating how different sectors are affected by carbon-futures implied uncertainty shocks, we uncover a clear green–brown asymmetry. Climate transition uncertainty shocks lower aggregate activity but their effects are unevenly distributed: carbon-intensive (brown) sectors such as oil, gas, and utilities experience deeper and more persistent contractions, while low-emission and technology-intensive (green) sectors display greater resilience and faster recovery. In a financial perspective, the price–dividend ratio falls for all sectors immediately after the shock but rebounds earlier and overshoots in the case of green industries, indicating a reallocation of investors' wealth toward transition-resilient assets. To assess whether these sectoral effects reflect broader financial volatility, we compare the carbon-futures shock with a recursively identified VIX shock, scaled to produce the same initial increase in aggregate volatility. The carbon-futures shock produces a larger gap in valuations between green and brown sectors and a deeper decline in brown-sector production. These differences suggest that the proxy captures a distinct dimension of risk not fully explained by the benchmark financial uncertainty shock.

Moving beyond this broad decomposition, a full sector-by-sector analysis shows that these asymmetries become even sharper. Once we examine individual industries, we find a clear gradient: sectors characterized by higher carbon intensity exhibit systematically larger declines in both output and valuations, whereas lower-emission sectors experience milder effects. This granular evidence extends the green–brown findings, demonstrating that the severity of climate uncertainty-induced adjustments increases monotonically with emission intensity.

Overall, this evidence supports the interpretation of carbon futures implied climate transition uncertainty as a distinct and market-based form of uncertainty that links policy ambiguity to macro-financial volatility. Our results highlight the importance of transparent and credible climate policy in reducing transition costs, stabilizing financial markets, and supporting the efficient allocation of capital toward low-carbon activities.

The remainder of the paper is structured as follows. Section 2 introduces the carbon futures realized variance-based climate uncertainty instrument, detailing its construction and validation. Section 3 presents the data and methodology. Section 4 reports the aggregate macro-financial results, whose robustness across specifications is discussed in Section 5. Section 6 explores sectoral heterogeneity at two complementary levels: a broad green–brown decomposition and a granular sector-by-sector analysis of real and financial responses, including their mapping to EPA emission intensities. Section 7 compares the carbon-futures shock with a generic financial-volatility shock. Section 8 concludes.

2 Complementary Perspectives on Climate Uncertainty

A key motivation of this paper is to complement the existing climate-policy uncertainty literature with a market-based perspective. Textual and news-based indicators have provided important insights into the evolution, salience, and intensity of climate-policy debates and have become widely used tools for studying uncertainty surrounding climate regulation. Financial markets, however, offer an additional source of information by revealing how investors assess and price transition-related risks. Because the two approaches capture different but related dimensions of the same

phenomenon, they should be viewed as complementary rather than competing measures.[2]

The realized-volatility innovations extracted from carbon futures prices provide a market-based indicator of climate-policy uncertainty. Since carbon allowance prices incorporate information about the expected trajectory, credibility, and timing of climate policies, fluctuations in their volatility reflect real-time reassessments of transition-related risks. This perspective is closely related to the proxy-SVAR literature on monetary policy, which emphasizes the informational content of financial market prices and the use of external instruments to identify exogenous shocks in typical policymaking environments characterized by communication, forward guidance, and information effects (see, e.g., Gertler and Karadi (2015), Nakamura and Steinsson (2018), and Miranda-Agrippino and Ricco (2021)).

Our approach therefore differs from studies relying primarily on textual indicators, such as Gavriilidis (2021), Ardia et al. (2023), and Känzig et al. (2025). Rather than focusing on the evolution of climate-policy discussions, our measure captures how uncertainty is reflected in asset prices and financial conditions. Driven by their trading profit-seeking, financial investors are likely incorporate information about prospective regulatory changes as soon as it affects investors' perceptions of transition risks, even before such developments become fully reflected in media coverage or public policy debates. As such, we propose that our measure may be particularly useful for studying the financial transmission of climate-policy uncertainty and its effects on macroeconomic and sectoral outcomes.

A Market-Based Proxy for Climate Policy Uncertainty

The realized volatility estimator derived from carbon futures prices is the core instrument of our empirical identification strategy. This estimator is *non-parametric, consistent under mild assumptions, and robust to model misspecification*, providing an ex-post measure of total return variation that efficiently aggregates high-frequency information from market prices (Andersen et al., 2001).

To implement this approach, we use data from the ICE U.S. Carbon Futures Index, which focuses specifically on U.S. carbon markets and aggregates futures prices from the two major regional cap-and-trade systems: the Western Climate Initiative (California and Québec) and the Regional Greenhouse Gas Initiative (Northeastern and mid-Atlantic US states).[3] The index tracks December futures on California and RGGI emission allowances, weighted by trading volume and rolled into the following year's contracts each autumn (ICE Data Indices, 2022). California's program, linked to Qu'ebec's, covers several sectors, including industry, power generation and transport fuels, while RGGI focuses on power generation in participating north-eastern states (International Carbon Action Partnership, 2025). Despite their limited emissions coverage, these markets represent economically important regions and offer insight into expectations about future climate policy.[4]

In the United States, the carbon futures markets are organized around cap-and-trade programs

[2] Cascaldi-Garcia et al. (2023) show that the interpretation of uncertainty shocks depends importantly on the construction of the underlying measure. Different indicators may capture distinct dimensions of uncertainty and need not map one-for-one into the same underlying economic mechanism.

[3] These contracts are standardized, fully collateralized, and centrally cleared, which lowers counterparty risk and creates a transparent price discovery process.

[4] In this respect, prices in these markets may contain useful information about climate-policy expectations and risk assessments, much as benchmark markets in other asset classes are often used to gauge broader developments beyond the activity directly covered by the market itself.

that regulate greenhouse-gas emissions at the state and regional level. Under these schemes, regulators set an upper limit on total emissions and issue a fixed number of allowances, each granting the right to emit one metric ton of CO_2-equivalent. Firms that emit less than their allocation can sell surplus allowances, while those exceeding their limits must purchase additional permits to comply with regulation. This creates a secondary market in which prices are determined by the interaction of demand—driven by firms' expected emissions, energy demand, and technological efficiency—and supply, which depends on the regulatory cap, allowance auctions, and policy adjustments.

Futures contracts on the allowances issued according to the Western Climate and the Regional Greenhouse Gas Initiatives are actively traded on the Intercontinental Exchange, where they serve both compliance entities and financial investors. Compliance participants use futures to hedge against the risk of higher carbon prices, while speculators and institutional investors provide liquidity and express views on the future path of policy and energy markets. As a result, futures prices embed the collective expectations of market participants regarding policy stringency, regulatory credibility, and compliance costs. Because these expectations are continuously updated as new political and regulatory information arrives, volatility in carbon futures prices provides a direct and high-frequency measure of how markets reassess climate-policy risk in real time. This feature makes ICE carbon futures the ideal source of our market-based proxy of climate-policy uncertainty.

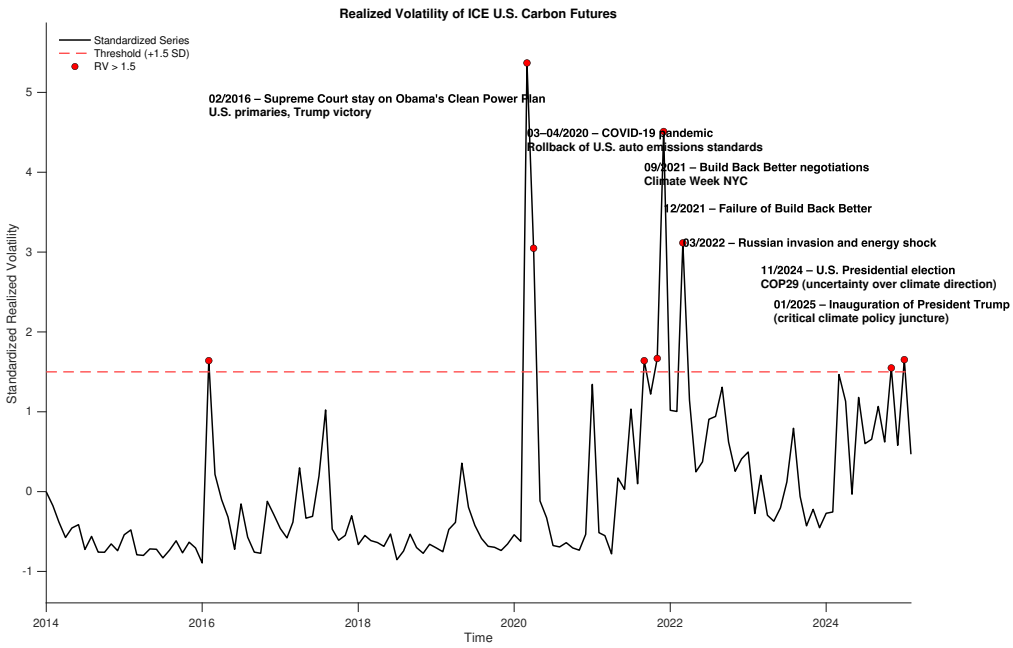


Figure 1
Realized Volatility of ICE U.S. Carbon Futures. Standardized realized volatility of the ICE U.S. Carbon Futures Index (2014–2025). Red dots indicate volatility spikes exceeding 1.5 standard deviations, corresponding to major climate-policy events.

Let P_t denote the daily closing price of the ICE carbon futures contract. Daily log-returns are

computed as $r_t \equiv \log P_t - \log P_{t-1}$, and monthly realized volatility as

$$RV_m = \sqrt{\frac{30}{n_m} \sum_{t=1}^{n_m} r_t^2},$$

where n_m is the number of trading days in month m, and the factor 30 adjusts for unequal trading days during a month. Figure 1 plots the standardized realized volatility measure for the ICE U.S. Carbon Futures Index. Visibly, the resulting time series displays pronounced spikes corresponding to episodes of elevated uncertainty about the credibility and the timing of U.S. climate policy.

The spike of 2016, for instance, followed the U.S. Supreme Court's suspension of the Clean Power Plan and coincided with the presidential election campaign, when the prospect of a Trump administration created significant uncertainty over the future course of U.S. climate policy. The 2020 episode overlapped with the rollback of vehicle emissions standards, both of which weakened perceptions of regulatory commitment to decarbonization. The fluctuations in carbon futures price realized volatility observed in 2021–22 reflect the turbulence surrounding the Build Back Better negotiations and the geopolitical shock of the Russian invasion of Ukraine, events that together heightened doubts about the pace and credibility of the energy transition.[5] Finally, the final spikes in late 2024 and early 2025 coincide with renewed political uncertainty linked to the U.S. presidential election, the COP29 negotiations, and the return of Donald Trump to the presidency. These are all developments that are likely to have intensified investors' concerns about the continuity of climate policy commitments. Financial markets appear to have reacted to these events through increased volatility andwidespread portfolio re-shuffling, which is consistent with on-going reassessment of policy credibility by market participants (see Monasterolo et al. (2019)).

In addition, following Miranda-Agrippino and Ricco (2021) and a broader finance literature which documents a degree of persistence in realized-volatility dynamics reflecting the slow adjustment of expectations to the arrival of new information, we extract the innovation component of realized volatility to construct an exogenous instrument to identify climate uncertainty shocks. Specifically, we remove the autoregressive portion of the series, isolating its true surprise element, consistent with the gradual assimilation of information by market participants, an established feature of imperfect-information environments (see Coibion and Gorodnichenko (2015)). To model this predictable component, we examine the stochastic properties of realized volatility by fitting alternative ARMA(p,q) specifications and selecting from them using the Bayesian and Hannan–Quinn information criteria. Both criteria select a parsimonious AR(1) representation:

$$RV_t = \alpha + \beta RV_{t-1} + \varepsilon_t.$$

This model is estimated over the full sample for carbon futures prices, January 2014 - February 2025. The residuals ε_t represent the realized-volatility innovations, that is, the unexpected component capturing real-time revisions in market expectations about climate policy, which provides an exogenous source of variation to support structural identification.

The choice of the ICE U.S. Carbon Futures as our benchmark instrument also reflects its

[5] The Build Back Better Act was a major climate-oriented policy proposal designed to expand clean-energy investment through large federal tax credits and support for low-carbon technologies. Its repeated legislative breakdowns in 2021–22 created substantial uncertainty about the future strength of U.S. climate policy, generating volatility in expectations about long-run decarbonization incentives.

superior first-stage performance relative to alternative carbon-market-based surprises that could be used for identification, as documented in Table 1. All alternative proxies are based on futures contracts traded on ICE and sourced from Bloomberg, and belong to the ICE Carbon Futures Index Family, covering the Western Climate Initiative (California–Québec), the Regional Greenhouse Gas Initiative (U.S. Northeast), their U.S. aggregation, and their global aggregation.[6] Moreover, all candidate proxies are constructed using the same realized-volatility methodology and correspond to innovations extracted from an autoregressive process of order one, selected according to standard information criteria, as in the case of the ICE U.S. index.

Instrument	Market coverage	First-stage F-statistic
ICE U.S. Carbon Futures (US aggregate)	WCI + RGGI	17.26
ICE California (WCI)	CA + QC	15.46
ICE RGGI	U.S. Northeast	0.92
Global Carbon Futures Index	International markets	3.46

Table 1: First-stage relevance of alternative carbon-market-based instruments. The table reports first-stage F-statistics from the projection of reduced-form VAR residuals on alternative carbon-market-based instruments. The proxied variable is the VIX. The BVAR includes VIX, IP, PCE, PD ratio, and FFR. Sample: February 2014–January 2025.

Beyond its conceptual appeal, the ICE U.S. Carbon Futures Index provides the strongest first-stage relevance for aggregate financial volatility among the alternative proxies considered, reflecting its broader market coverage and tighter link to U.S. financial conditions. Its first-stage F-statistic is comfortably above the conventional threshold of 10 typically recommended in the literature (Stock and Watson, 2018; Känzig et al., 2025), whereas proxies based on more regional or global carbon markets display substantially weaker relevance for the U.S. financial channel.

To assess the climate-policy content of our measure, we first examine the correlations between monthly carbon-futures volatility innovations and other uncertainty indicators (Table 2). We then conduct a daily event study around 167 climate-policy events, using absolute daily carbon-futures returns to measure volatility.

Variables	RV/Climate uncertainty shocks	RVUS	CPU Index	VIX
RV/Climate uncertainty shocks	1.000	0.849	0.273	0.306
RVUS	0.849	1.000	0.413	0.445
CPU Index	0.273	0.413	1.000	0.280
VIX	0.306	0.445	0.280	1.000

Table 2: Correlation matrix between uncertainty indicators. Correlation matrix is estimated with reference to the sample period February 2014–January 2025. The matrix concerns correlations between realized-volatility innovations (RV/climate uncertainty shocks), the realized volatility of ICE U.S. Carbon Futures (RVUS), the textual Climate Policy Uncertainty (CPU) index of Gavriilidis (2021), and the VIX implied-volatility index.

The strongest correlation with RV/climate uncertainty shocks occurs in the case of RVUS (0.85), as the innovation is derived from the same realized-volatility series. This likely reflects the me-

[6] Further details on the ICE Carbon Futures Index Family are available at https://www.ice.com/fixed-income-data-services/index-solutions/commodity-indices/carbon-futures.

chanical link between the two measures: the autoregressive filtering may remove the persistent component of volatility, while the remaining innovation captures the policy-related uncertainty behind unexpected movements in the carbon futures market. The correlation of RV/climate uncertainty shocks with the textual CPU index (0.27) is moderate, suggesting that our market-based proxy may capture a component of climate-policy uncertainty that is related to, but distinct from, the narrative-based measure derived from textual analysis. Finally, the correlation with the VIX (0.31) provides evidence of a financial transmission channel linking climate-policy uncertainty to broader market volatility. This connection indicates that carbon-market shocks are internalized by investors and transmitted to aggregate financial conditions through standard volatility channels. The moderate size of this correlation is consistent with a climate-specific source of risk that interacts with, but does not merely replicate, general financial uncertainty.

Overall, these correlations support the interpretation of the realized-volatility innovation as a meaningful proxy for climate-policy uncertainty. It shows sufficient association with existing uncertainty indicators to ensure empirical relevance, while remaining distinct enough to avoid its redundancy. This balance between overlap and specificity reinforces the suitability of RV/climate uncertainty shocks as an external instrument for identifying exogenous climate policy uncertainty shocks in a Bayesian Proxy-SVAR framework.

Event-Study Validation of the Instrument

The correlations in Table 2 show that our proxy is related to other measures of uncertainty, but not whether it reacts to climate-policy news. We check this with a daily event study on the ICE U.S. Carbon Futures Index. The event list, which contains 167 events, is built from official sources (Appendix Table 10) and covers four types of news. The first covers announcements of the outcomes of quarterly RGGI and California–Qu'ebec allowance auctions. The second covers decisions directly affecting the two regional programs, including changes to emissions limits and regulations, states joining or leaving the programs, and court rulings on their legal status. The third covers federal and international climate-policy developments, including EPA rules on power-plant emissions and related court decisions, congressional debates on climate legislation, and changes in U.S. participation in the Paris Agreement. The fourth is presidential and midterm elections and the climate-policy executive actions of each new administration.

We measure volatility with the absolute daily log return, $|r_t|$, which is the high-frequency counterpart of the realized volatility behind our instrument. Because both trading activity and the typical size of daily price changes grow substantially over the sample, we normalise each event by its own pre-event benchmark. Specifically, we divide $|r_t|$ by the average absolute return over the 60 trading days ending one week before the event, so that the benchmark is not contaminated by anticipation of the news, and a value of one corresponds to the pre-event level of volatility. Figure 2 reports the average response across all climate-policy events. Volatility is in line with its benchmark level in the two days before the news, rises to about twice that level on the day of the news, and then returns towards it.[7]

Figure 3 decomposes this average by type of event. Allowance auctions produce a pronounced

[7]The average over the event day and the day after differs from one with a t-statistic of 5.5. This results are similar across benchmark windows. They also hold when we exclude the COVID-19 months or the months of Russia's invasion of Ukraine. Section 5.3 examines the sensitivity of the structural estimates to these two episodes.

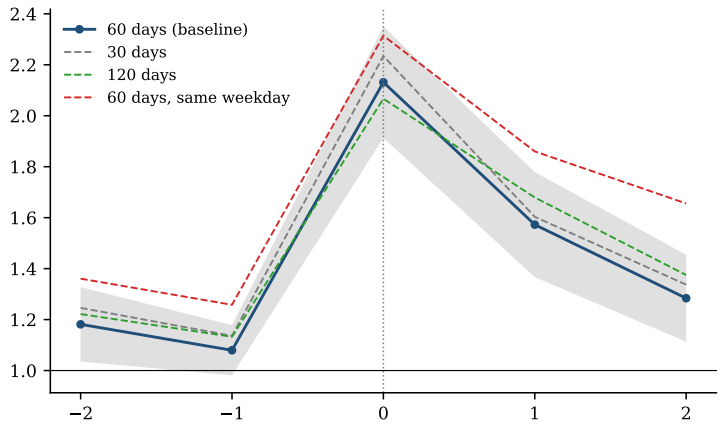


Figure 2
Carbon-Futures Volatility around U.S. Climate-Policy Events. Absolute daily return of the ICE U.S. Carbon Futures Index in the trading days around an event (day 0), relative to its pre-event average. Solid line: baseline, with ±1 standard error band. Dashed lines: alternative benchmarks.

reaction: when the clearing price becomes known, volatility is two and a half times its pre-event level (panel a). Decisions on the two regional programs raise volatility on the event day and even more on the following day (panel b), while federal and international policy, which affects these markets only indirectly, produces a more moderate response (panel c). Finally, we consider the two elections of Donald Trump, which reversed the direction of federal climate policy: on the first trading day after the 2016 and 2024 votes, volatility was 18 and 4 times its pre-election level (panel d).

These evidence shows that the volatility of carbon futures responds to news about climate policy at both the regional and the federal level. It responds to developments specific to the two regional markets, such as auction outcomes and changes in their rules and membership. It also reflects turning points in federal climate policy, such as the regulation of power-plant emissions and the related court rulings, and presidential elections, whose outcome determines the direction of federal policy.

Taken together, the evidence supports the use of the proxy as an instrument: it is relevant for aggregate financial volatility (Table 1), it is related to, but distinct from, existing measures of uncertainty (Table 2), and it responds to climate-policy news (Figures 2 and 3).

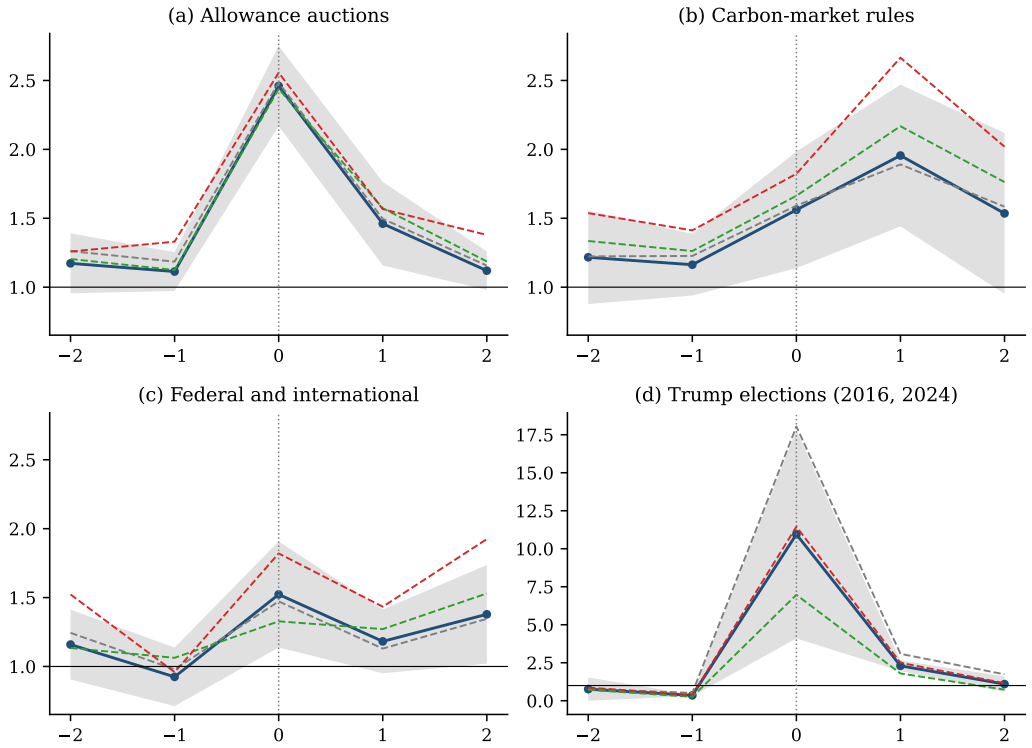


Figure 3
Carbon-Futures Volatility around Climate-Policy Events, by Type of Event. Absolute daily return of the ICE U.S. Carbon Futures Index in the trading days around an event (day 0), relative to its pre-event average. Solid line: baseline, with ±1 standard error band. Dashed lines: alternative benchmarks. Number of events: 88 (a), 39 (b), 25 (c), 2 (d).

3 Data and Methodology

3.1 Data

Our empirical analysis relies on a monthly dataset combining standard U.S. macro-financial indicators with a market-based measure of climate policy uncertainty. The sample spans the period February 2014 - January 2025. The data include variables capturing financial conditions, real activity, besides our high-frequency proxy for climate-related financial uncertainty.

The CBOE Volatility Index (VIX) is included in the analysis as the benchmark measure of aggregate financial uncertainty. Widely regarded as the "fear gauge" of the U.S. equity markets, it reflects investors' expectations of near-term volatility implied by S&P 500 options. Following the large literature on uncertainty and macro-financial dynamics (Bloom, 2009; Baker et al., 2016), the VIX serves as a reference measure of market-based uncertainty that allows us to assess how climate-related shocks interact with broader financial volatility.

Our proxy for climate policy uncertainty is based on the realized volatility of the ICE U.S. Carbon Futures Index, obtained from Bloomberg. The macroeconomic variables are sourced from FRED database: the VIX, the Industrial Production (IP) index, the Personal Consumption Expenditure (PCE) price index, and the Federal Funds Rate (FFR). The S&P 500 price–dividend ratio (PD) is from Robert Shiller's dataset.[8] We focus on the price–dividend ratio rather than the S&P 500 index itself, as it provides a valuation-based measure that filters out short-term price fluctua-

[8]Shiller's data are available at https://shillerdata.com/.

tions and better captures investors' expectations about future cash flows and discount rates.[9] This approach aligns with the extensive literature showing that valuation ratios summarize investors' long-horizon expectations and are informative about long-run risks, long-run growth uncertainty, and discount-rate variation, see Philips (1999).[10]

3.2 Bayesian Proxy-SVAR Framework

We estimate a Bayesian Structural VAR(p) model identified through an external instrument. Let y_t denote an $n\times 1$ vector of de-meaned, endogenous variables following the reduced-form representation

$$y_t = B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma_u),$$

and the corresponding structural form

$$A_0 y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t, \qquad E[\varepsilon_t \varepsilon_t'] = I_n,$$

where $u_t = A_0^{-1}\varepsilon_t$ and A_0^{-1} maps structural shocks into reduced-form innovations.

Identification of the structural shock of interest, denoted ε_t^k, follows the proxy BVAR approach of Stock and Watson (2018) and Mertens and Ravn (2013), applied to uncertainty shocks by Carriero et al. (2015). The method exploits an external instrument z_t that satisfies the following relevance and orthogonality conditions:

$$E[\varepsilon_t^k z_t] \neq 0,$$
$$E[\varepsilon_t^j z_t] = 0 \quad \forall\, j \neq k.$$

The first condition (relevance) ensures that z_t is correlated with the targeted structural innovation, while the second (exogeneity) requires independence from all other contemporaneous shocks. Together, these conditions allow the covariance between the VAR residuals and the proxy to isolate the structural disturbance of interest, because they imply that the reduced-form shock most strongly associated with z_t must coincide with ϵ_t^k and hence can be recovered without imposing further structural restrictions.

The BVAR is estimated following Miranda-Agrippino and Ricco (2021) and using standard Normal–Inverse Wishart priors; the priors' tightness is set as in Giannone et al. (2015). Posterior inference is performed via Gibbs sampling, allowing full propagation of parameter and identification uncertainty to the posterior distribution of the structural parameters. As explained earlier, the model includes VIX, IP, PCE, PD, and FFR. All variables enter the VAR in natural logarithms except for FFR. Given the relatively short time span of investigated sample, the lag order is set to $p = 6$, allowing the model to capture short-run propagation without an excessive loss of degrees of freedom. The use of six lags in a monthly VAR is not unusual (see, for example, Ludvigson et al.

[9] The price–dividend ratio is a standard forward-looking valuation measure in asset pricing. In a standard log-linear Campbell- Shiller decomposition (see Campbell and Shiller (1988)), the PD ratio encodes expectations of future dividend growth and discount rates, whereas index prices alone commingle these components with short-term noise. Because climate-related uncertainty affects both expected long-run cash flows (via growth prospects, damages, stranded-asset risk, (see Campbell and Shiller (1998) and Giglio et al. (2021)) and discount rates (via risk and ambiguity premia), PD movements provide a cleaner lens through which to study how climate-related shocks transmit to valuations.

[10] Further details and descriptive statistics of the dataset are provided in Appendix A

(2021), Caggiano et al. (2021), Cascaldi-Garcia and Galvao (2021)).

Given posterior draws of (B, Σ_u), identification proceeds by projecting the reduced-form innovations u_t onto the external instrument $z_t = \varepsilon_t^{RV}$, which corresponds to the innovation in the realized volatility of ICE carbon futures log-price changes. This step recovers the contemporaneous impact vector associated with the structural shock, from which the posterior distributions of A_0^{-1} is then derived. The relevance of the instrument arises from the fact that carbon allowance markets incorporate new information about the expected trajectory of climate regulation at high frequency, making realized volatility a direct proxy for financial uncertainty related to climate-policy news. In practice, we assess instrument relevance using the first-stage F-statistic from the projection of the reduced-form residuals on the realized-volatility instrument. The statistic equals 17.26, providing evidence of relevance,providing evidence of instrument relevance. This evidence is complemented by the event study in Section 2 which supports the climate-policy content of the underlying carbon-futures volatility measure. Identification additionally requires the instrument to be uncorrelated with other contemporaneous structural shocks. To assess the sensitivity of our estimates to potential confounding factors, we repeat the analysis using an instrument residualized on selected carbon-market, energy, policy and financial variables (Section 5).

4 Main Results

Figure 4 reports the impulse responses of all endogenous variables in the VAR to a climate transition uncertainty shock normalized to raise the VIX by one standard deviation, corresponding to an increase of 30.65 percent. IRFs are reported in percentage points.

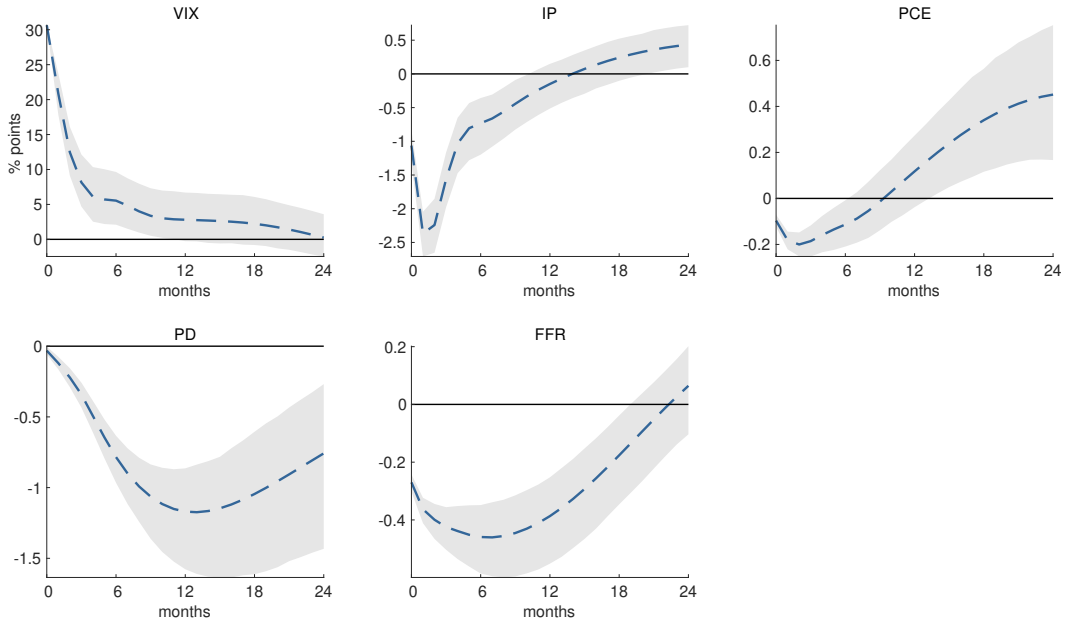


Figure 4
The Effects of Climate Policy Uncertainty Shocks Impulse responses to the identified climate transition uncertainty shock for all endogenous VAR variables (VIX, IP, PCE, PD ratio, and FFR). The shock is identified using the realized volatility innovation from ICE carbon futures log-price changes as the external instrument. Shaded areas denote 68% posterior credible intervals. IRFs are reported in percentage points. Sample: February 2014–January 2025.

To facilitate the interpretation of the dynamic responses shown in Figure 4, Table 3 reports a

concise quantitative summary of their main features, including impact effects, peak responses, and the horizon over which each response remains statistically significant.

Overall, the results show that an unanticipated increase in climate-related financial uncertainty generates macro-financial effects comparable to those typically associated with tightening uncertainty shocks. The VIX rises sharply on impact and remains elevated in the short and medium run, confirming the activation of a strong financial volatility channel. At the same time, real activity and asset valuations decline in the months following the shock. Industrial production contracts significantly, falling by about 1 percent on impact and reaching a trough of approximately -2.4 percentage points after one to two months. It remains significantly below baseline for roughly ten months before gradually recovering. The PCE price index initially falls, reaching a trough of about -0.20 percentage points and remaining significantly below baseline for about six months. It then recovers and rises above baseline, reaching +0.45 percentage points at the end of the two-year horizon. Asset valuations respond strongly and persistently. The price-dividend ratio declines steadily over time, reaching a maximum contraction of about -1.17 percentage points after roughly one year, and remains significantly below its pre-shock level throughout the two-year horizon. This persistence points to a prolonged adjustment in market valuations, consistent with sustained increases in discount rates and risk premia. Monetary policy reacts accommodatively. The federal funds rate declines on impact by about 27 basis points and continues to fall gradually, reaching a maximum reduction of approximately 46 basis points after seven months, before slowly converging back toward baseline. This response suggests a systematic accommodative monetary policy reaction to the weaker macroeconomic and financial conditions induced by heightened climate uncertainty.

Variable	Impact	Peak Response	Persistence (Month)
VIX	30.65	30.65	11
IP	-1.06	-2.37	10
PCE	-0.10	-0.20	6
PD	-0.03	-1.17	24
FFR	-0.27	-0.46	18

Table 3: Quantitative Summary of Aggregate Impulse Responses to Climate Policy Uncertainty Shocks. The table reports impact and peak impulse responses corresponding to a Climate Policy Uncertainty Shock. Peak responses are defined as the maximum absolute response. The last column reports the final horizon at which the response remains statistically significant at 68%. IRFs are reported in percentage points. Sample: February 2014–January 2025.

The short-run fall in prices suggests that our proxy primarily captures the *second-moment* component of policy uncertainty rather than the *first-moment* policy shock itself. Indeed, the existing literature generally finds that climate-policy shocks that raise carbon prices are inflationary, as tighter environmental regulation increases firms' production costs, constrains energy supply, and generates cost-push pressures. Within the EU Emissions Trading System, both Känzig (2023) and Hensel et al. (2024) show that carbon price policy shocks are inflationary and contractionary, as higher allowance prices raise firms' marginal costs and depress output, confirming the supply-side nature of these disturbances. Our measure, however, captures fluctuations in the volatility of the rate of change of futures prices rather than in their levels. Hence, while changes in the carbon price reflect *policy actions* or expectations of regulatory tightening, movements in volatility reflect *uncertainty* surrounding the timing, credibility, and intensity of such actions. This interpretation

is further supported by the strong response of the VIX to the identified shock, since the VIX is widely regarded as a canonical second-moment measure of uncertainty.

The initial fall in prices and the speed of the real effects are both consistent with this reading. When the path of regulation becomes harder to predict, firms postpone investment and hiring and households increase precautionary saving (Bernanke, 1983; Bloom, 2009); with nominal rigidities, the resulting fall in demand lowers output and prices together (Leduc and Liu, 2016; Basu and Bundick, 2017). Since postponing a decision takes effect immediately, activity falls within a few months and recovers as postponed decisions are carried out, which is the pattern we find for industrial production. The medium-run rise of the price level, from roughly the tenth month onwards, goes instead in the direction of the inflationary response to carbon-price shocks.

Interestingly, the responses of financial variables—such as the price–dividend ratio, and the policy rate—are considerably more persistent than those of real variables like industrial production and consumer prices, with statistically significant effects extending up to roughly 18 months for the policy rate, and the entire horizon for asset valuations, compared to about 6–10 months for the initial decline in prices and output (Table 3). This pattern is consistent with a financial transmission mechanism through which climate-policy uncertainty primarily operates via risk re-assessment, asset re-pricing, and adjustments in financial conditions, thereby amplifying macro-financial fluctuations and inducing a sustained easing in policy rates.

5 Robustness Checks

5.1 Assessing Instrument Exogeneity

As discussed in Section 3, the exogeneity condition requires that the realized-volatility innovation be unrelated to structural shocks other than the one of interest. A natural concern is that carbon-futures volatility also reflects changes in the level of carbon prices, developments in energy markets, general policy uncertainty, financial-market conditions, or the outcome of presidential elections. To address it, we follow Miranda-Agrippino and Ricco (2021) and residualize the instrument on variables capturing these factors.We regress the realized-volatility innovation on the monthly log return of the ICE U.S. Carbon Futures Index, monthly log changes in the Henry Hub natural gas price and the Economic Policy Uncertainty index of Baker et al. (2016), the lagged log VIX, and a dummy for presidential election months. We then use the residuals as the external instrument.[11]

Figure 5 compares the impulse responses obtained with the baseline and residualized instruments. Their direction, timing and persistence are similar, although the residualized instrument produces somewhat larger effects: industrial production falls by 3.1 percent at the trough, compared with 2.4 percent in the baseline. These findings suggest that the baseline results are not driven by variation in the instrument associated with the selected controls.

5.2 Alternative Instruments and Robustness of Identification

We complement the baseline specification with additional exercises that consider alternative instruments and related identification checks. First, we use WCI-based surprises as an alternative

[11]The natural gas price is the Henry Hub spot price from FRED. The regression is estimated over the same sample as the baseline, February 2014–January 2025.

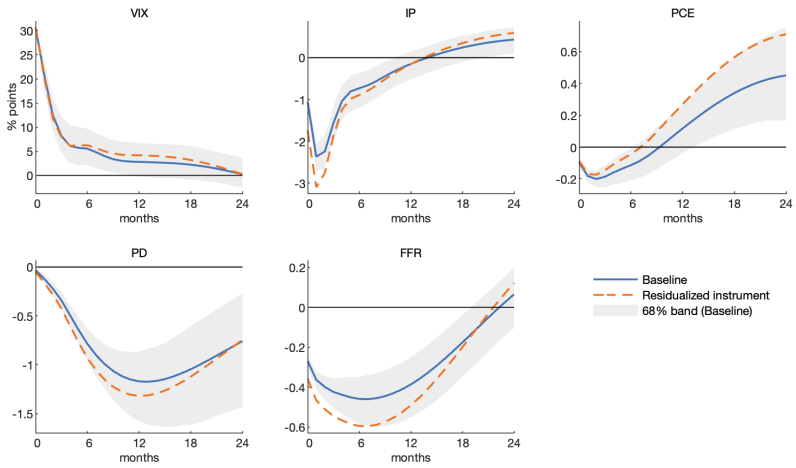


Figure 5
Robustness: Residualized Instrument. Impulse responses to the identified climate uncertainty shock when the external instrument is purged of the carbon-futures return, the change in natural gas prices, the change in the U.S. Economic Policy Uncertainty index, the lagged VIX and presidential-election months. Shaded areas denote 68% posterior credible intervals for the baseline specification. IRFs are reported in percentage points. Sample: February 2014–January 2025.

external instrument. Second, we allow for richer dynamics in the construction of the same ICE-based realized-volatility surprise used earlier by estimating AR(2) and ARMA(1,1) specifications instead of the baseline AR(1). In all cases, the corresponding instruments satisfy the standard relevance condition, with first-stage F-statistics above conventional thresholds. These exercises test whether our results depend on the choice of proxy (WCI vs US aggregate) or on the dynamic filter used to extract innovations (AR(1), AR(2), ARMA(1,1)).

The associated impulse responses are reported in Appendix C (Figures 22–24). Results remain quantitatively and qualitatively unchanged across specifications, confirming that our findings are robust to alternative market-based instruments and different autoregressive structures used to extract the unexpected component of realized volatility.

5.3 Sensitivity to the COVID-19 and Ukraine Episodes

Our sample includes the outbreak of COVID-19 and Russia's invasion of Ukraine. Although both episodes involved uncertainty about climate policy, they also generated major macroeconomic disruptions that may have affected the volatility of the carbon future through other channels. In addition, in spring 2020, California–Qu'ebec allowance prices fell sharply and auction demand weakened. In the same weeks a federal court ruled on the legality of the California–Québec linkage and the federal government rolled back vehicle emission standards. The invasion of Ukraine, in turn, raised doubts about the pace and the priorities of the energy transition.

Despite their climate-policy relevance, we exclude these episodes from the instrument to assess whether our findings depend on the unusually large disruptions that accompanied them. Specifically, we treat March–May 2020 and February–March 2022 as missing observations of the instrument, while estimating the VAR over the full sample. Figure 6 compares the resulting impulse responses with the baseline and shows that their sign and shape remain broadly unchanged.[12]

[12]The excluded months are treated as missing observations of the instrument; the VAR is estimated over the full

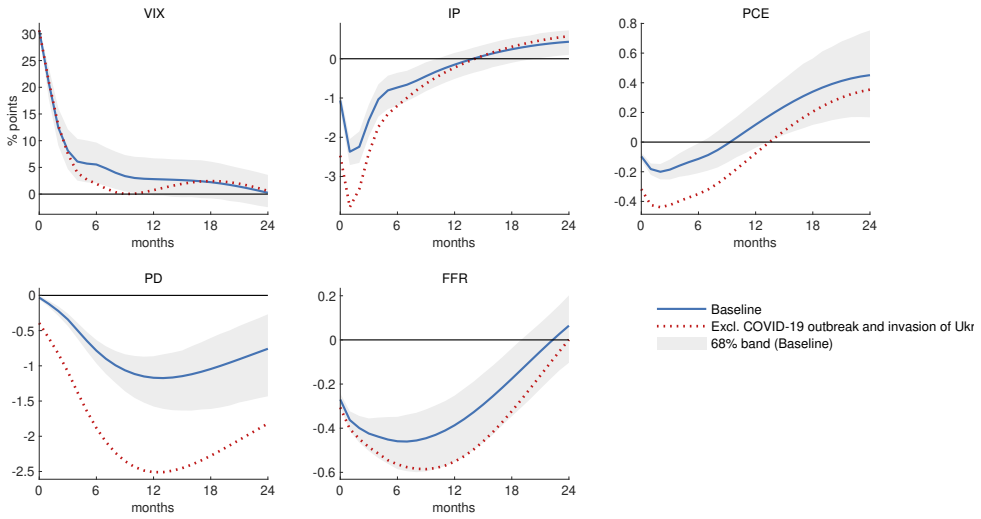


Figure 6
Robustness: Excluding the COVID-19 and Ukraine Episodes. Impulse responses to the identified climate uncertainty shock in the baseline (solid line) and when the external instrument excludes March–May 2020 and February–March 2022 (dotted line). Shaded areas denote 68% posterior credible intervals for the baseline specification. IRFs are reported in percentage points. Sample: February 2014–January 2025.

5.4 Alternative Model Specifications

In this section, we assess the robustness of our baseline findings to alternative model specifications and sample definitions. The purpose of these exercises is to assess whether the dynamic responses to carbon futures–implied uncertainty shocks are sensitive to alternative modeling assumptions and estimation choices.

We complement the BVAR approach with an alternative estimation framework based on local projections, following Ramey (2016). She argues that LPs allow the estimation of impulse responses directly at each horizon, reducing sensitivity to potential misspecifications in the VAR dynamics. Third, we implement a Bayesian LP approach following Miranda-Agrippino and Ricco (2021) who have proposed to regularize horizon-specific LP coefficients by centering their prior around the iterated coefficients from the baseline BVAR, thereby combining the efficiency of VAR estimation with the robustness of LP methods. The prior tightness is estimated at each horizon within a hierarchical Bayesian framework, allowing the data to determine the optimal degree of shrinkage.[13]

Our findings from the first robustness exercise consisting of the adoption of alternative lag lengths. The outcomes from the alternative model specifications, including the LP and BLP estimations, are shown in Figure 7. Across these exercises, the dynamic responses display remarkably similar magnitudes and persistence vs. those shown in our section on main empirical results, confirming that the estimated effects of carbon futures-implied uncertainty shocks are not driven by model-specific assumptions.

We have also verified that the results are robust to alternative lag-length choices (see Appendix B). In addition, the findings remain unchanged when varying prior hyperparameters and shrinkage settings in the BVAR specification; full results are available upon request.

sample.
[13]For the BLP the pre-sample period, running from February 2014 to February 2018.

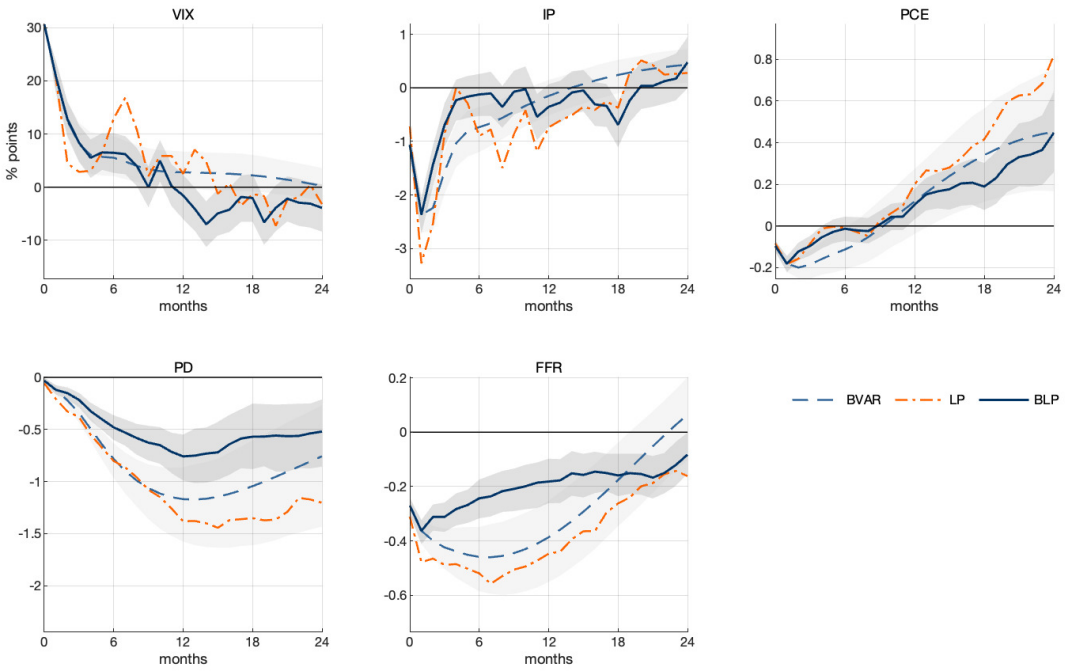


Figure 7
BVAR vs LP or Bayesian LP: Robustness to Different Model Specifications. Impulse responses to the identified climate uncertainty shock for all endogenous variables (VIX, IP, PCE, PD ratio, and FFR). The shock is identified using the realized-volatility innovation from ICE carbon futures as the external instrument. Shaded areas denote 68% posterior credible intervals.IRFs are reported in percentage points. Sample: February 2014–January 2025.

5.5 Alternative Variables: S&P 500 and Unemployment

To further assess the robustness of our findings, we re-estimate the BVAR using alternative aggregate variables that capture the same macro-financial phenomena, yet through different indicators. Specifically, we first replace the log-price–dividend ratio with the S&P 500 log-index; we also substitute industrial production with the US unemployment rate.[14] Figure 8 shows the results obtained when the S&P 500 index replaces the price–dividend ratio in the VAR specification. The overall pattern of responses remains nearly identical to the baseline: the VIX rises immediately, while the S&P 500 index declines on impact, reflecting a tightening of financial conditions. The magnitude of these responses is very similar to those obtained with the price–dividend ratio in Figure 4; however, the negative effect on the market valuation index turns out to be less persistent, and in this specification the price level remains below baseline over the whole horizon, without the medium-run increase observed in the baseline.

Figure 9 shows that replacing industrial production with the unemployment rate does not alter the recessionary dynamics illustratd earlier. Following a climate transition uncertainty shock, the unemployment rate rises persistently, mirroring the contraction in output observed in the baseline set of results. The similarity of the responses across specifications reinforces the robustness of our findings and indicates that the transmission mechanism of climate uncertainty shocks is stable across different indicators of real/financial activity.

[14]The S&P 500 index is obtained from Robert Shiller's website, while the US unemployment rate is sourced from the FRED database.

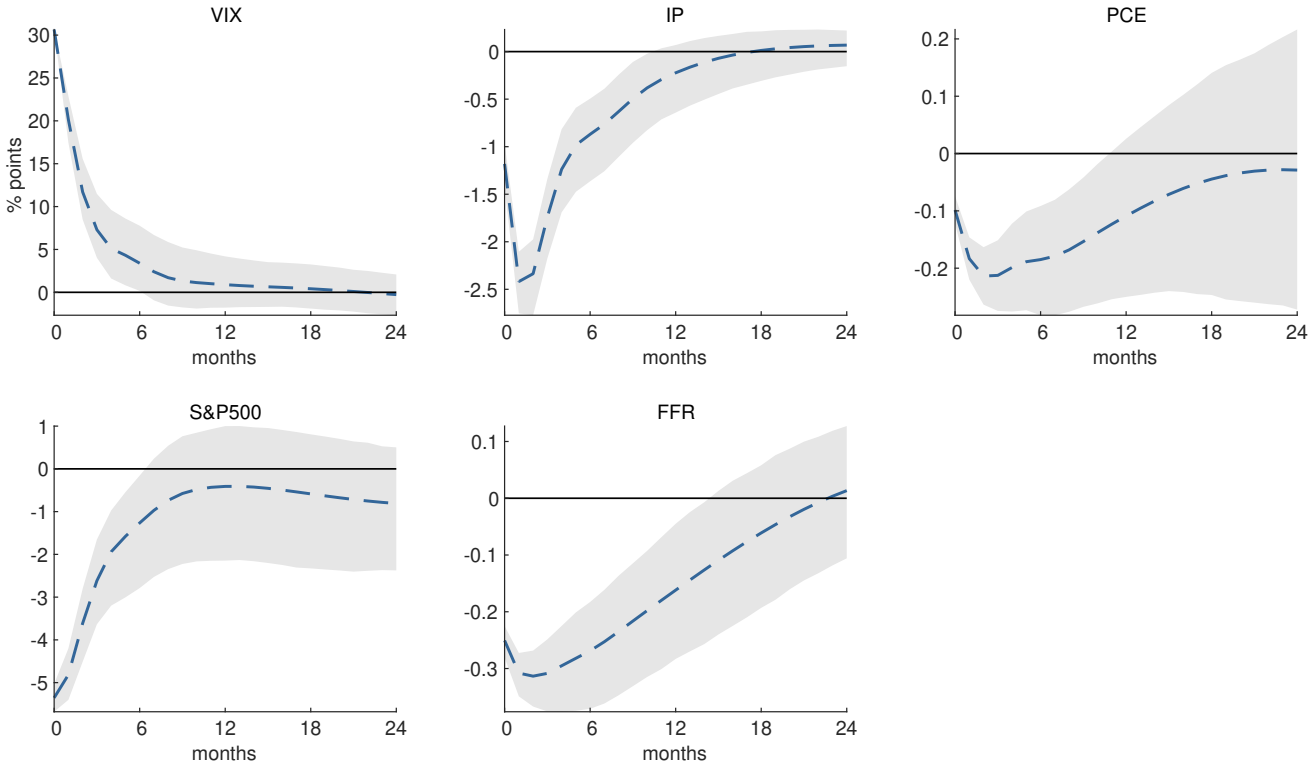


Figure 8
Alternative Aggregate: S&P 500 Index. Impulse response functions to the identified climate uncertainty shock when the S&P 500 index replaces the price–dividend ratio in the Bayesian Proxy-SVAR. The specification includes VIX, industrial production, PCE, S&P 500, and the federal funds rate. Shaded areas denote 68% posterior credible intervals. IRFs are reported in percentage points. Sample: February 2014–January 2025.

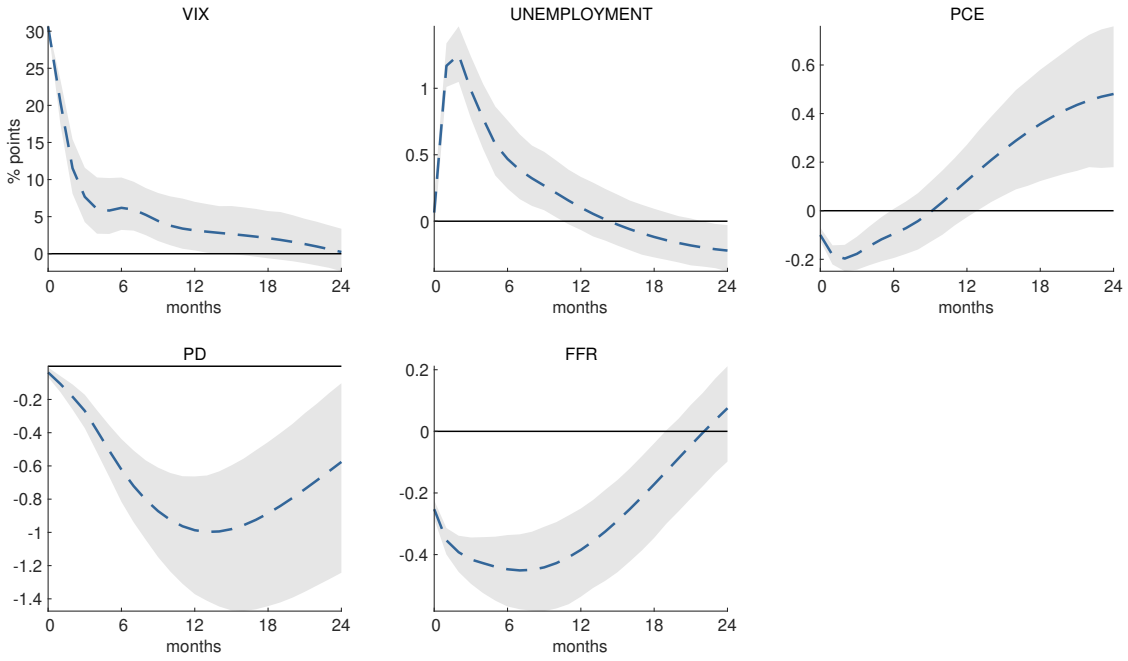


Figure 9
Alternative Aggregate: Unemployment Rate. Impulse response functions to the identified climate uncertainty shock when the unemployment rate replaces industrial production in the Bayesian Proxy-SVAR. The specification includes VIX, unemployment rate, PCE, price–dividend ratio, and the federal funds rate. Shaded areas denote 68% posterior credible intervals. IRFs are reported in percentage points. Sample: February 2014–January 2025.

5.6 Controlling for General Policy and Energy Market Conditions

A natural concern is that carbon-futures volatility may co-move with broader policy uncertainty or energy-market conditions. To assess whether our results are driven by these factors, we augment the baseline Bayesian Proxy-SVAR with additional controls capturing (i) general economic policy uncertainty and (ii) oil market dynamics.

First, we include the U.S. Economic Policy Uncertainty (EPU) index of Baker et al. (2016), which captures broad policy-related uncertainty beyond the climate domain. Second, we augment the VAR with the logarithm of the WTI crude oil price (sourced from FRED), thereby controlling for energy-market conditions that may influence both carbon markets and macroeconomic outcomes.

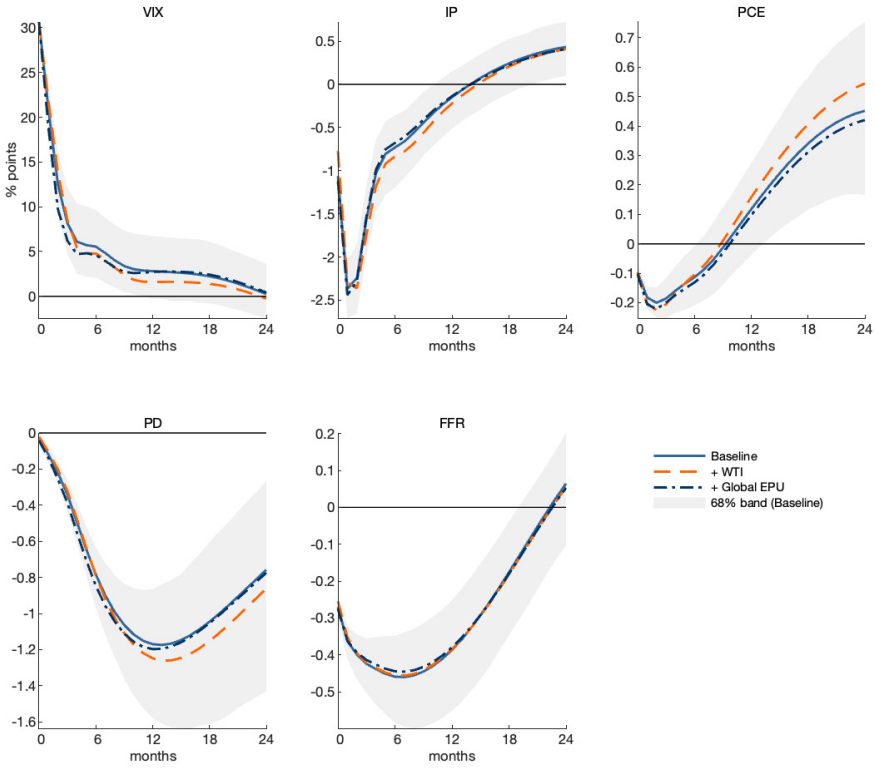


Figure 10
Robustness: Controlling for U.S. EPU and Oil Prices. Impulse responses to the identified climate uncertainty shock when augmenting the VAR with the U.S. Economic Policy Uncertainty index and the log WTI crude oil price. The specification includes VIX, IP, PCE, PD ratio, FFR, and the additional control variable. Shaded areas denote 68% posterior credible intervals. IRFs are reported in percentage points. Sample: February 2014–January 2025.

Figure 10 reports the resulting impulse responses, which remain quantitatively and qualitatively similar to the baseline specification. These results indicate that the main macro-financial effects we document are robust to controlling for broader policy uncertainty and oil-market conditions.

6 Analysis of the Heterogeneity in Green vs. Brown Sectoral Responses

6.1 Sectoral Heterogeneity: Green vs. Brown Industries

To investigate how climate uncertainty shocks propagate throughout the economy, we construct broad *green* and *brown* aggregates for both real activity (industrial production) and asset valuations (price–dividend ratio). This sectoral perspective is crucial because the transition to a low-carbon economy entails an uneven redistribution of risks and investment opportunities across industries. While climate-related policies may penalize carbon-intensive activities through tighter regulations or higher abatement costs, they can simultaneously benefit green and technology-oriented sectors by stimulating innovation and investment in sustainable production. In our macro-finance sectoral setting, we define as *brown* the sectors that are fossil-fuel-based, to include the energy supply industries (e.g., utilities, oil and gas extraction, and petroleum refining) that are directly exposed to transition and regulatory risks. Conversely, the *green* sectors comprise low emission and transition enabling manufacturing activities, such as electrical equipment and electronic components, which provide the technological backbone for electrification, energy efficiency, and renewable integration.

6.1.1 Sectoral Industrial Production Analysis

We use the sectoral industrial production indices made available by the Federal Reserve's G.17 data repository and based on the North American Industry Classification System (NAICS) classification.[15] We construct two broad aggregates reflecting industries' exposure to climate-transition risk: *brown* (carbon-intensive) and *green* (low-emission or transition-enabling) sectors. The brown sectors–*Utilities*, *Oil and Gas Extraction*, and *Petroleum and Coal Products*–represent the core of the fossil-fuel value chain. They exhibit high direct greenhouse gas emissions, rely heavily on carbon-based energy inputs, and are most exposed to policy shocks such as carbon pricing and emission standards. This grouping is consistent with the literature which identifies fossil energy production, utilities, and energy-intensive manufacturing as the most vulnerable to climate policy risk (Battiston et al., 2017; Bolton and Kacperczyk, 2021; Känzig et al., 2025). In contrast, *Computers and Electronic Products* and *Electrical Equipment, Appliances, and Components* are categorized as green sectors. These industries exhibit low operational emissions and act as key transition enablers by providing essential inputs for electrification, renewable-energy systems, and energy-efficiency technologies, such as semiconductors, sensors, electric motors, and power-conversion components. This grouping is consistent with the literature, which identifies technology-intensive and innovation-driven manufacturing sectors as central to the low-carbon transition and as beneficiaries of climate-related and policy support (Vona et al., 2018; Alessi et al., 2024; Zhang et al., 2024). In practice, we construct the green and brown industrial production indices as simple (equal-weighted) averages of the NAICS industrial production series included in each group. This aggregation provides the data

[15] NAICS is the standardized taxonomy used by U.S., Canadian, and Mexican statistical agencies to categorize all establishments into economically meaningful and mutually exclusive industries. NAICS replaced the older Standard Industrial Classification (SIC) codes by grouping firms according to similarity in production processes rather than end products. This has yielded a more consistent mapping between technological input structure and measured output. In the context of the G.17 release, NAICS codes determine the aggregation used to construct the industrial production series: establishment-level output is first assigned to detailed 4–6-digit NAICS industries, then aggregated hierarchically into higher-level sectors.

useful to estimate a summary measure of the average responses of high- vs. low-emission sectors to climate uncertainty shocks while preserving comparability with the aggregate VAR specification.

We therefore extend our baseline analysis by re-estimating the BVAR to include sectoral activity data. The vector of endogenous variables now comprises the VIX, the aggregate IP, PCE, PD, FFR, and the two sectoral, overall IP indices for the green and brown industries, respectively. Apart from this sectoral IP extension, the model is estimated using the same data, sample period, priors, and identification strategy as in the baseline specification. In fact, the impulse response functions of the aggregate variables, VIX, IP, PCE, PD, and FFR, are generally unchanged vs. those obtained in the main model.[16]

As shown in Figure 11, industrial production in the brown sectors experiences a deeper and more persistent contraction relative to the green sectors, with a peak decline of around 1.90% compared to 0.97% for green industries. Panel (b) reports the difference between the two responses, computed draw by draw. It is positive at all horizons, peaking at about 1 percentage point after eight months. This asymmetric reaction reflects the greater exposure of carbon-intensive industries to transition risk, consistent with standard theoretical predictions by which high emissions or technologically inflexible production face disproportionately larger valuation losses when the probability of future carbon regulation increases. This occurs because firms with higher emissions intensities bear greater adjustment costs—through stranded assets, higher future compliance expenditures, or accelerated depreciation of brown capital—which amplifies their sensitivity to shocks in policy stringency or expectations about decarbonization pathways (Dietz et al. (2016); Lemoine and Traeger (2016); Barnett et al. (2020); Engle et al. (2020); Van der Ploeg and Rezai (2020); Bolton and Kacperczyk (2021)).[17]

6.1.2 Sectoral Price–Dividend Ratios Analysis

After documenting these asymmetric responses in real activity, we next examine whether climate-policy uncertainty affects financial valuations across sectors. To this end, we construct sectoral PD ratios from the Fama–French 49 Industry portfolios.[18] The classification mirrors the one used for industrial production: brown industries include *Oil (Petroleum and Natural Gas)* and *Utilities (Electric and Gas)*, while the green industries include *Chips (Electronic Components)* and *Electrical Equipment*. Throughout the sectoral valuation analysis we only use portfolios that contain at least ten firms in every month of the sample, so that the implied dividend series are not driven by the payout decisions of a handful of firms; this requirement excludes the *Coal* portfolio, which counts as few as four firms over the sample.[19] Each sectoral PD ratio is computed using the previous twelve months of implied dividends (annualized), and series are aggregated up at sectoral level as equally-weighted means.[20]

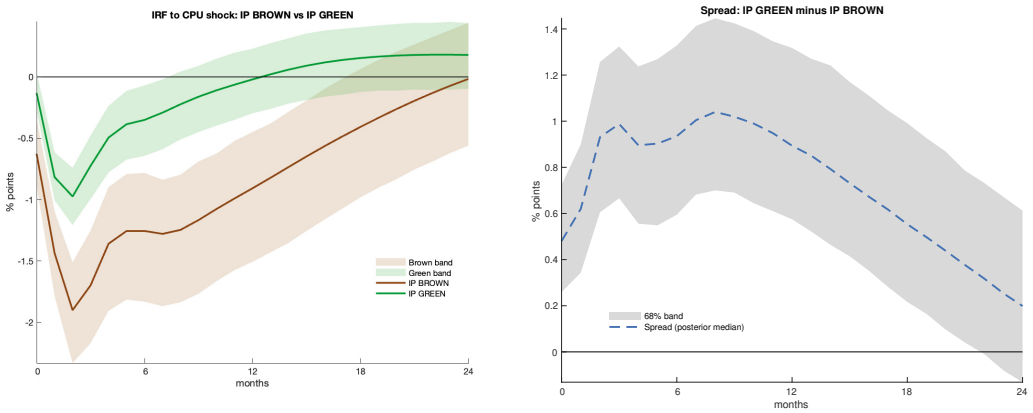
[16] Detailed results and IRF plots are available from the authors upon request.

[17] Equilibrium frameworks demonstrate that carbon-intensive sectors load more heavily on the transition component of climate risk because investors require compensation for holding assets whose cash flows depend on regulatory and technological trajectories that are both uncertain and downward-skewed for polluting industries (see Pástor et al. (2022); Giglio et al. (2021).

[18] Stock prices and dividends are sourced from French's website.

[19] The number of firms in each portfolio is taken from Kenneth French's data library. The same requirement is applied to the sector-by-sector analysis.

[20] The Fama–French industry portfolios are based on the SIC classification, which is the same system used for industrial production data before NAICS replaced it in 1997. This means that the sectoral mappings in the PD analysis follow the SIC structure and therefore line up with the older industrial production categories.

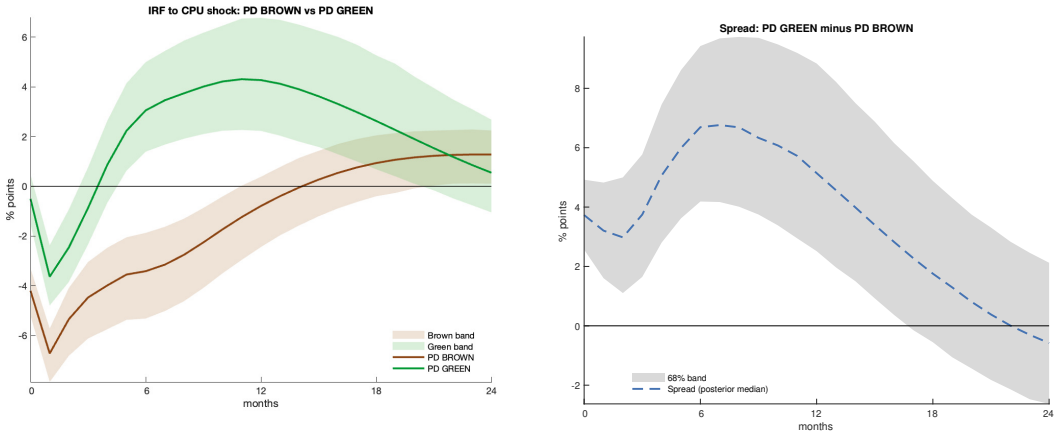


(a) Responses **(b)** Difference, green minus brown

Figure 11
Sectoral Industrial Production Responses to a Climate Uncertainty Shock. Panel (a): impulse responses of the green and brown industrial production indices in the Bayesian Proxy-SVAR that includes the VIX, aggregate IP, PCE, PD, FFR and the two sectoral indices. Sectoral indices are equal-weighted averages of the constituent NAICS series: *Brown* = Utilities (2211–2212), Oil and Gas Extraction (211), Petroleum and Coal Products (324); *Green* = Computers and Electronic Products (334), Electrical Equipment (335). Shaded areas denote 68% posterior credible intervals. Panel (b): difference between the green and the brown response, computed draw by draw within the same VAR; IRFs in percentage points. Sample: February 2014–January 2025.

As in earlier analysis, we re-estimate the BVAR including sectoral PD ratios for green and brown industries, using the same specification, the same data and sample period, and identification strategy as in the baseline model. Figure 12 shows that climate uncertainty shocks initially depress asset valuations in both groups, consistent with a tightening of financial conditions and an increase in risk premia. However, over intermediate horizons the PD ratio of green sectors rebounds quickly and overshoots its pre-shock level, while brown-sector valuations fall sharply in the months after the shock and return to their pre-shock level only after about a year. The green–brown difference is positive and significant at the 68 percent level over intermediate horizons (Panel (b)). This pattern suggests that heightened climate-policy uncertainty triggers a temporary flight to safety away from all stocks issued in all sectors and a subsequent reallocation of wealth that reflects a shift in investor preferences toward greener industries and away from brown.

This mechanism is consistent with asset pricing models in which forward-looking investors revise expected cash flows and discount rates asymmetrically across sectors when climate transitions uncertainty rises (see Pástor et al. (2022)): green firms benefit from stronger optionality on future regulation and lower expected compliance costs, whereas carbon-intensive firms face greater downside risk due to the potential tightening of emissions constraints (Lemoine and Traeger (2016); Barnett et al. (2020); Bolton and Kacperczyk (2023)). Empirically, studies show that periods of elevated climate uncertainty are associated with declines in brown asset valuations and relative inflows into low-carbon or high ESG score-tilted assets, reflecting both risk-based rebalancing and evolving investor beliefs about long-term transition pathways (see, e.g., Ceccarelli et al. (2024); Cornelli et al. (2025). Moreover, evidence from green–brown return spreads corroborates the idea that markets tend to reward firms perceived to be better aligned with expected future policy regimes during episodes of regulatory uncertainty, reinforcing the asymmetric dynamics observed in the PD ratios (see, e.g., Basaglia et al. (2025)).



(a) Responses **(b)** Difference, green minus brown

Figure 12
Sectoral Price–Dividend Ratio Responses to a Climate Uncertainty Shock. Panel (a): impulse responses of the green and brown sectoral price–dividend ratios in the Bayesian Proxy-SVAR that includes VIX, IP, PCE, PD, FFR and the two sectoral indices. Sectoral indices are equal-weighted averages of the constituent SIC groups: *Brown* = Oil (30), Utilities (31); *Green* = Chips (37), ElcEq (22). Shaded areas denote 68% posterior credible intervals. Panel (b): difference between the green and the brown response, computed draw by draw within the same VAR; dashed line: posterior median; shaded areas: 68% credible intervals. IRFs in percentage points. Sample: February 2014–January 2025.

Overall, these results reveal a pronounced *green–brown asymmetry* in both real and financial responses to climate uncertainty shocks. While climate-related uncertainty depresses aggregate activity, its effects are unevenly distributed across sectors: carbon-intensive industries experience deeper and more persistent contractions, whereas low-carbon and technology-oriented sectors display greater resilience and faster recovery. The estimated impulse responses indicate that a rise in climate-related financial uncertainty reduces overall economic activity, with stronger and longer-lasting effects on brown industries. In terms of market valuations, the price–dividend ratio initially declines across all sectors following a climate shock, but subsequently rebounds for green industries, suggesting a reallocation of investors' wealth towards transition-resilient assets as the economy adjusts. These findings support the view that the asymmetric responses of the two sectors are particularly important, because they highlight how climate transition shocks are likely to redistribute risks and investment opportunities across the economy, accelerating capital flows towards greener industries while exposing carbon-intensive activities to higher financial vulnerability. Such heterogeneity across sectors aligns with firm-level evidence showing that firms with greater exposure to climate and transition risks tend to react more negatively to increases in climate-policy uncertainty (Känzig et al., 2025; Basaglia et al., 2025).

6.2 Granular Sectoral Heterogeneity

The previous green vs. brown analysis already highlighted meaningful asymmetries in how broad groups of industries respond to climate transition risk shocks. To obtain a more granular and comprehensive picture of how climate policy uncertainty propagates across the economy, we extend the analysis to the full cross-section of U.S. industries. Specifically, we disaggregate real activity using the complete set of 23 NAICS industrial production series from the Federal Reserve's G.17 database, and we examine sectoral valuation dynamics using a wide sub-set of the 49 Fama–French

industry portfolios. This finer sector-by-sector approach uncovers pronounced differences in both the magnitude and the persistence of real and financial responses across U.S. industries—patterns that are largely obscured in more aggregated analyses—and crucially allows us to match each sector to its corresponding EPA supply-chain emission intensity, thereby linking observed sectoral responses to their underlying carbon exposure.

6.2.1 Sectoral Disaggregation: Industrial Production and Emission-Based Heterogeneity

In this subsection, we estimate a separate Bayesian VAR for each of the 23 industrial production sectors reported in the Federal Reserve's G.17 release, following the official NAICS industry-group classification. Each sectoral VAR has exactly the same specification adopted in the main model, with the only difference being the inclusion of one sectoral IP index at a time. This ensures that the identification and dynamic structure remain fully comparable across models and that the sectoral setup closely mimics the propagation of the same aggregate climate uncertainty shock. Identification is based on the *same* external instrument used in the aggregate Proxy-SVAR—namely, the innovation to the realized volatility of the ICE U.S. Carbon Futures Index. This setup guarantees that differences across sectors reflect genuine heterogeneity in the estimated transmission mechanism, rather than differences in the way shocks are identified and constructed or model specification.

Table 4 reports the mean impulse responses of each sector over horizons of 0–24 months. The results reveal substantial heterogeneity. Carbon-intensive and energy-related activities exhibit the sharpest output contractions. Oil and Gas Extraction displays the largest decline in IP, with an average response of –1.51%, followed by Motor Vehicles and Parts (–0.96%), Coal Mining (–0.85%), and Natural Gas Distribution (–0.81%). These sectors are characterized by high reliance on fossil-fuel inputs, technologically rigid production processes, and strong exposure to climate- and regulatory-driven demand shifts. Other energy-intensive industries, such as Printing, Fabricated Metal Products, Petroleum and Coal Products, and Plastics and Rubber, also respond negatively to climate transition risk shocks but with lower magnitudes.

In contrast, several lower-emission manufacturing sectors exhibit muted or slightly positive responses, such as Computer and Electronic Products, Machinery, Electrical Equipment, Wood Products and Textiles. Figure 13 provides a visual ranking of all sectors by their exposure to climate transition uncertainty shocks, highlighting the systematic concentration of negative responses among fossil-fuel-based or energy-intensive activities (colored in brown). Even among sectors that are not overtly fossil-fuel-based, the responses appear ordered in a way that correlates closely with their indirect energy dependence, capital intensity, and position within upstream supply chains. Industries such as Printing, Fabricated Metal Products, Aerospace, and Plastics and Rubber Products display moderately negative responses, consistent with their indirect reliance on energy-intensive inputs and carbon-exposed supply chains. By contrast, more innovation-oriented or less energy-dependent sectors—such as Computer and Electronic Products, Machinery, Electrical Equipment, or Furniture—show negligible responses, underscoring the role of technological flexibility, lower emissions intensity, and greater adaptability to regulatory shifts.

IP Sector (NAICS)	Mean IRF (0–24)
Oil and Gas Extraction (211)	-0.01506
Motor Vehicles and Parts (3361–3)	-0.00955
Coal Mining (2121)	-0.00845
Natural Gas Distribution (2212)	-0.00811
Printing and Related Support (323)	-0.00723
Fabricated Metal Products (332)	-0.00685
Petroleum and Coal Products (324)	-0.00535
Aerospace and Misc. Transport (3364–9)	-0.00417
Plastics and Rubber Products (326)	-0.00340
Computer and Electronic Products (334)	-0.00180
Electric Power Generation (2211)	-0.00158
Furniture (337)	-0.00152
Machinery (333)	-0.00074
Nonmetallic Mineral Products (327)	-0.00052
Electrical Equipment (335)	-0.00039
Primary Metals (331)	-0.00033
Paper (322)	-0.00027
Food, Beverage, and Tobacco (311–2)	0.00062
Apparel and Leather (315–6)	0.00147
Wood Products (321)	0.00148
Misc. Durable Goods (339)	0.00244
Textiles (313–4)	0.00320
Chemicals (325)	0.00366

Table 4: Mean IRF of Industrial Production Sectors to a climate uncertainty Shock (0–24 Months). The Mean IRFs represent the average response over 24 horizons, estimated from a sector-specific VAR. The sectoral VAR includes VIX, aggregate IP, PCE, PD, FFR, and the sectoral IP index. Sample: February 2014–January 2025.

Linking climate uncertainty Sensitivity to IP Emission Intensity

To explain the cross-sectional pattern of responses uncovered earlier, we match each sector with its supply-chain greenhouse gas emission intensity using the U.S. Environmental Protection Agency's harmonized dataset of NAICS-level emission factors. These data provide total GHG intensity (in kg CO_2e per 2022 USD of output) at a granular 6-digit level.[21] Since industrial production sectors are defined at a coarser NAICS aggregation, sector-level emissions are computed as averages of the corresponding 6-digit emission factors.[22]

Table 5 reports the resulting emission-intensity ranking. A clear alignment emerges between the highest-emission industries (e.g., Coal Mining, Natural Gas Distribution, Oil and Gas Extraction, Petroleum and Coal Products, Plastics and Rubber) and the sectors most negatively exposed to carbon futures-implied uncertainty shocks, with Paper and Chemicals as the main exceptions.

To develop a clearer statistical understanding of the relationship between emission intensity and the real effects of climate uncertainty shocks, we plot the sectoral mean IRFs against their corresponding emission measures. Figure 14 provides a visual illustration of this link: sectors with higher greenhouse-gas intensities tend to cluster in the lower part of the scatter, indicating more adverse output responses to a climate shock. For estimation purposes, emission intensities are rescaled by dividing by 1000 (so that one unit corresponds now to one kg of CO_2e per USD), which improves numerical comparability in the subsequent regression. The blue line in the figure depicts

[21] EPA Supply-Chain Greenhouse Gas Emission Factors by NAICS data are accessible through the U.S. Government Open Data Portal at this link.

[22] Electric Power Generation (NAICS 2211) is omitted due to missing EPA margin adjustments, those corrections that align industry-level emissions or energy-use data with economic activity measures by correcting for wholesale and retail trade margins, transportation margins, and other distributional markups embedded in the use of intermediate goods.

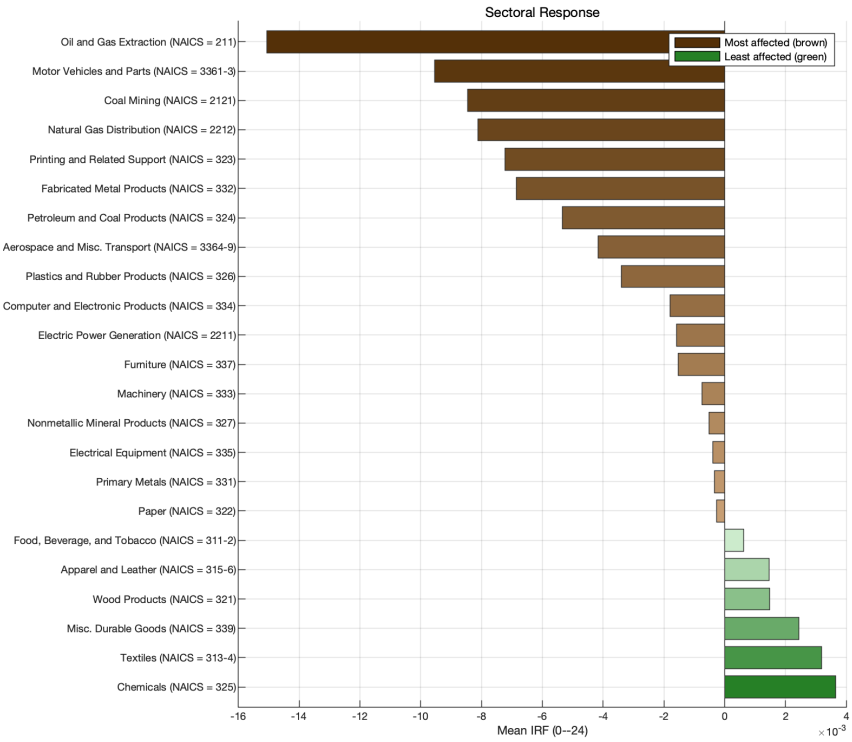


Figure 13
Industrial Production Sectoral Exposure to Climate Uncertainty Shocks. Ranking of IP sectoral mean impulse responses to a climate uncertainty shock. Brown- (Green-) coded sectors represent the most (least) negatively affected industries. The sectoral responses represent the average response over 24 horizons, estimated from a sector-specific VAR. Each sectoral VAR includes VIX, aggregate IP, PCE, PD, FFR, and the sectoral IP index. Sample: February 2014–January 2025.

the fitted regression line, which corresponds to the specification we formally discuss below. Regressing the sectoral mean IRFs on the scaled emission intensity yields an estimated slope coefficient of $\hat{\beta} = -0.0118$, statistically significant at the 10% level, confirming the negative relationship seen in the scatter plot: more (less) carbon-intensive sectors experience systematically more (less) adverse responses to climate transitionrisk shocks. Overall, the results support the idea that the climate shock proxy systematically hits "brown" sectors harder: higher-emission industries exhibit more negative IRF responses, with larger magnitudes and greater persistence over the forecast horizon. In contrast, lower-emission and technology-enabling sectors display milder reactions to the shock.

6.2.2 Sectoral Disaggregation: Price–Dividend Ratio and Emission-Based Heterogeneity

To complement the real-side analysis, we examine now the occurrence of any heterogeneous valuation effects across industries. We construct sectoral price–dividend ratios using the Fama–French 49 Industry Portfolios, based on stock returns and dividends from Kenneth French's data library.[23]We start from the 49 Fama–French industry portfolios. We drop the nine for which dividend histories are not available over the full sample, leaving 40 portfolios, and then the seven that do not have at

[23]Fama–French 49 Industry Portfolios, available through Kenneth French's Data Library at this link.

Sector (IP)	NAICS	Emissions (g CO2e per USD)
Coal Mining	2121	747.0
Natural Gas Distribution	2212	532.0
Paper	322	469.48
Oil and Gas Extraction	211	377.0
Plastics and Rubber Products	326	365.44
Primary Metal	331	336.08
Chemical	325	309.00
Petroleum and Coal Products	324	308.62
Nonmetallic Mineral Product	327	303.53
Food, Beverage, and Tobacco	311–2	302.85
Aerospace and Misc. Transport	3364–9	209.68
Printing	323	205.0
Fabricated Metal Products	332	203.34
Textiles and Products	313–4	198.00
Machinery	333	175.34
Motor Vehicles and Parts	3361–3	170.19
Wood Products	321	163.93
Electrical Equipment	335	145.45
Furniture	337	139.19
Miscellaneous Durable Goods	339	118.0
Apparel and Leather Goods	315–6	101.08
Computer and Electronic Products	334	51.10

Table 5: Emission Intensity by Sector (g CO_2e per USD). Sector-level emission intensities are obtained by matching each industry to the U.S. Environmental Protection Agency's harmonized dataset of NAICS-based supply-chain greenhouse gas emission factors, reported at the 6-digit level and provided without margins.

least ten firms in every month. The final sample contains 33 portfolios.[24]Importantly, the Fama–French industry portfolios are classified according to the historical SIC system, which underlies all portfolio groupings in this dataset.

We estimate a separate BVAR for each sector, following the same specification used in the industrial production analysis. The vector of the endogenous variables now includes the VIX, aggregate IP, PCE, PD, FFR, and, one at a time, the sectoral PD ratio.

Table 6 reports the mean impulse responses of the sectoral PD ratios to a climate uncertainty shock over the horizon 0–24 months. The results reveal substantial cross-sectional variation. Medical Equipment, Laboratory Equipment, and Real Estate exhibit the largest negative responses, followed by sectors such as Miscellaneous, Business Services, Oil, and Chemicals. In contrast, several industries show positive or strongly positive long-run responses to the shock, including Chips, Electrical Equipment, Retail, and Entertainment. Beyond the clear dispersion in mean PD responses, the ranking suggests a deeper economic structure. The sectors at the lower end of the distribution—Medical Equipment, Laboratory Equipment, Real Estate, Oil, and Chemicals—share features that amplify their sensitivity to uncertainty shocks, such as heavy reliance on physical capital, exposure to regulatory or input-cost risks, and dependence on long-duration cash flows. These characteristics make their valuations particularly responsive to shifts in discount-rate premia induced by climate-related uncertainty. Conversely, sectors with positive long-run PD responses tend to be more innovation-intensive (Chips, Electrical Equipment), more flexible in adjusting business models (Retail), or oriented toward discretionary consumption and digital content (Entertainment), all of which make them more resilient to transitional uncertainty or even attractive destinations for investor rebalancing.[25]

[24]The criterion excludes Agriculture, Beer & Liquor, Coal, Shipbuilding & Railroad, Shipping Containers, Textiles and Tobacco. The slope of the emission regression is negative for any threshold between 5 and 20 firms.

[25]This pattern is consistent with the view that climate uncertainty reshapes the relative attractiveness of economic

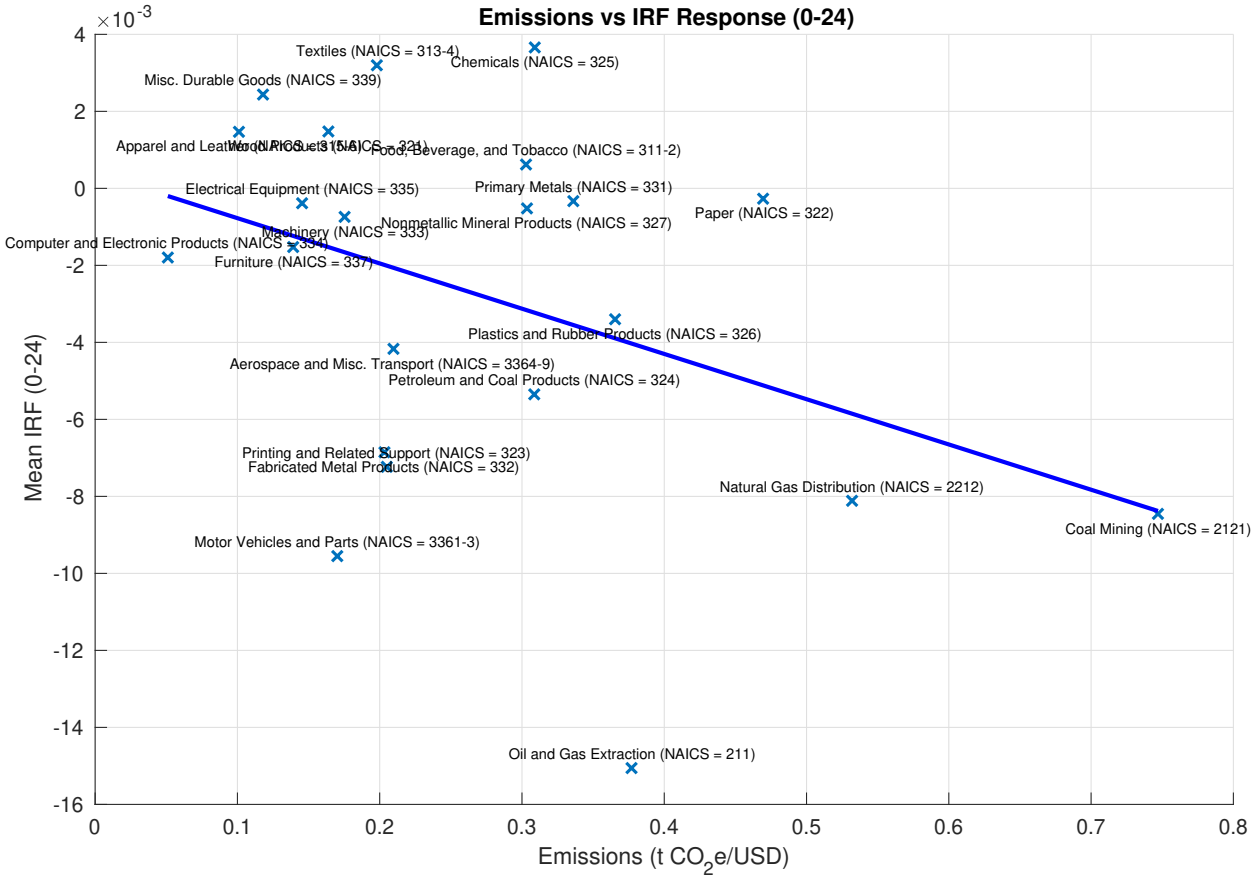


Figure 14
Emissions vs. Sectoral IP Responses. The scatter plot illustrates the existence of a negative correlation between emission intensity and average impulse responses to climate uncertainty shocks. The blue line represents the fitted regression line. Emission intensities are rescaled by dividing by 1000 ((so that one unit corresponds now to one kg of CO_2e per USD)).

Sector	Mean IRF	Sector	Mean IRF
Medical Equipment (MedEq)	-0.05428	Telecommunications (Telcm)	0.01089
Laboratory Equipment (LabEq)	-0.04019	Food Products (Food)	0.01135
Real Estate (RlEst)	-0.03770	Restaurants, Hotels & Motels (Meals)	0.01189
Miscellaneous (Other)	-0.02781	Wholesale (Whlsl)	0.01275
Business Services (BusSv)	-0.02416	Computers (Hardw)	0.01300
Petroleum & Natural Gas (Oil)	-0.02054	Toys/Recreation (Toys)	0.02031
Chemicals (Chems)	-0.01682	Retail (Rtail)	0.02121
Insurance (Insur)	-0.00914	Electronic Components (Chips)	0.02346
Utilities (Utilities)	-0.00895	Mining (Mines)	0.02406
Household Goods (Hshld)	0.00042	Electrical Equipment (ElcEq)	0.02493
Banking (Banks)	0.00265	Aircraft (Aero)	0.02894
Drugs (Drugs)	0.00314	Transportation (Trans)	0.03288
Building Materials (BldMt)	0.00479	Books/Printing (Books)	0.03338
Finance (Fin)	0.00490	Apparel (Clths)	0.03740
Steel Works (Steel)	0.00511	Automobiles & Trucks (Autos)	0.16231
Construction (Cnstr)	0.00585	Entertainment (Fun)	0.21018
Machinery (Mach)	0.00604		

Table 6: Mean IRF of Sectoral Price–Dividend Ratios to a Climate Uncertainty Shock (0–24 Months). The IRFs represent the average response over 24 horizons, estimated from a sector-specific VAR. Each sectoral VAR includes VIX, aggregate IP, PCE, PD, FFR, and the sectoral PD index. Sample: February 2014–January 2025. The "Miscellaneous" sector includes residual or unclassified industries, including niche utilities and small-scale service providers.

activities not only through direct emissions exposure, but also through broader channels such as technological

Emission Mapping and Valuation Responses

To examine whether the heterogeneity in valuation responses is related to transition risk, we map each SIC-based sector into its underlying NAICS industries and match these to the EPA supply-chain emission factors. Because the SIC and NAICS classification systems are not mutually consistent, the mapping follows the official SIC–NAICS crosswalk, which may associate a single SIC code with several NAICS industries and, conversely, link some NAICS codes back to multiple SIC categories.[26] This structural mismatch reflects the different classification logics underlying the two taxonomies. Such non-one-to-one correspondence naturally introduces cross-code variation in the resulting sector-level emission measures, as the NAICS industries linked to a given SIC sector may differ substantially in their emission intensities. To ensure that the mapped emission proxies remain coherent and comparable across sectors, we apply a log-dispersion filter. Specifically, we compute for each SIC sector the difference $\Delta_{\log} = \log(\max) - \log(\min)$ across all associated NAICS emission factors and retain only those sectors with internal dispersion below the threshold $\Delta_{\log} \leq 8.5$.[27]

Figure 15 plots the mean PD response against the corresponding emission intensity. As in the real-side analysis of the averaged IRFs for industrial production, the scatter plot reveals a clear downward pattern: higher-emission sectors tend to exhibit more negative valuation responses to climate uncertainty shocks. A simple regression of sectoral mean PD responses on rescaled emission intensity yields a slope coefficient of $\beta = -0.0623$, statistically significant at the 10% level. This confirms that carbon-intensive portfolios systematically experience more negative valuation effects in response to climate-policy uncertainty shocks. Importantly, the sensitivity of valuations to climate-policy uncertainty is not confined to a handful of extreme polluters but reflects a gradient visible throughout the cross-section. Although the most carbon-intensive sectors (Miscellaneous, Utilities, Oil) anchor the lower end of the distribution, the negative association extends across the entire emissions range. For instance, mid-emission manufacturing industries—such as Chemicals—also lie meaningfully below zero. This reinforces the idea that, under uncertainty about future climate regulation, it is difficult for carbon-intensive industries to generate valuation gains, even if they are not directly regulated in the short run. Investors appear to treat emissions intensity as a structural exposure that scales downside risk. Finally, the dispersion around the fitted line is asymmetric, with low-emission sectors showing more vertical spread, including several mildly positive responses. This suggests the presence of factors—such as technological adaptability, intangible capital, or expected demand shifts—that can offset the negative discount-rate effect for greener sectors but rarely do so for brown ones.

Overall, and similarly to the findings for IP responses, the evidence indicates that climate uncertainty shocks propagate asymmetrically across valuation measures: carbon-intensive industries experience larger and more persistent declines in PD ratios, while low-emission and transition-enabling sectors display milder or even positive valuation adjustments. This valuation asymmetry mirrors the real-side sectoral heterogeneity and reinforces the interpretation of carbon future-driven climate shocks as a disturbance that disproportionately affects brown industries.

Taken together, the granular results in this section show that emission intensity is a key driver of

adaptability, capital intensity, and sectoral demand elasticities.

[26] Available at NAICS crosswalk's link.

[27] This procedure leaves a final sample of 27 sectors for which the mapped emission intensity can reasonably be interpreted as a meaningful summary measure.

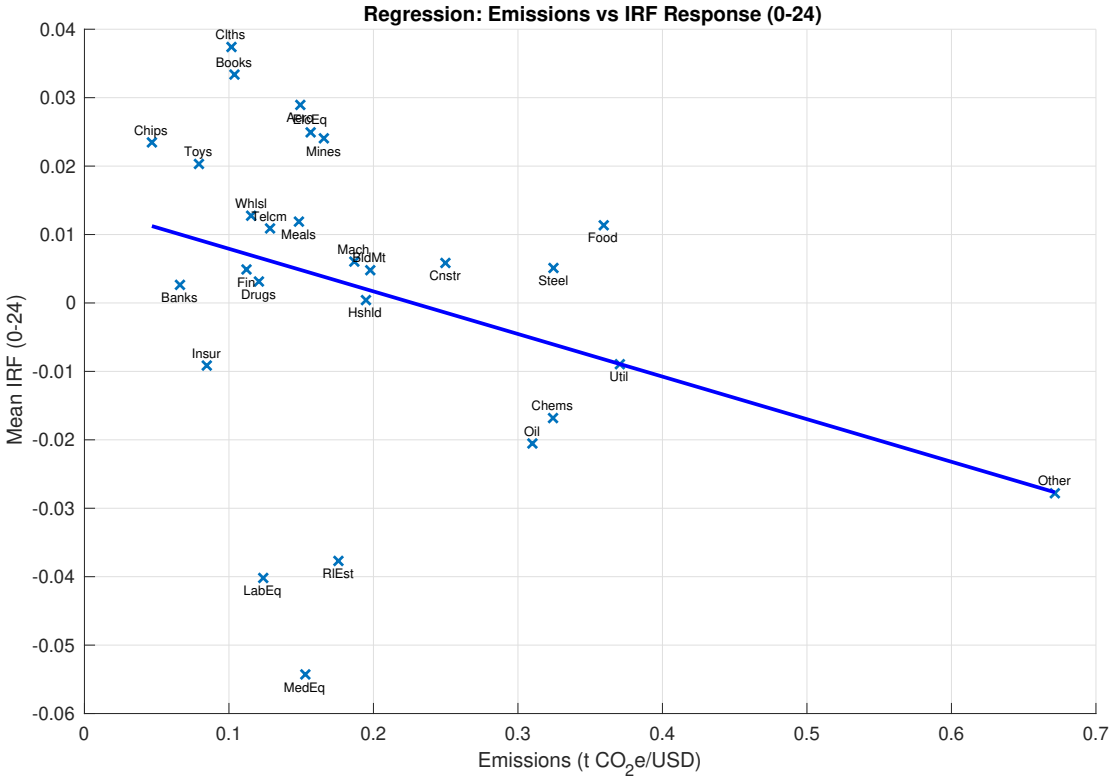


Figure 15
Emissions vs. Sectoral PD Responses. The scatter plot illustrates the existence of a negative correlation between emission intensity and average impulse responses to climate uncertainty shocks. The blue line represents the fitted regression line. Emission intensities are rescaled by dividing by 1000 (one unit equals one kg of CO_2e per USD).

sectoral vulnerability to climate shocks: industries with higher carbon intensity are systematically hit harder, both in terms of real activity and equity valuation losses. In addition, on the financial side, several low-emission sectors even display a positive valuation response in the form of higher price-dividend ratios. This is consistent with a reallocation of investor wealth toward industries perceived as more resilient or less exposed to transition risk (Pástor et al., 2022; Ardia et al., 2023).

6.3 Green Re-Pricing Mechanism

To gain additional insights, we next examine what the asymmetric adjustments reported above may imply for long-horizon valuation spreads across brown and green sectors. To analyse how climate-policy uncertainty is differentially prices across industries, we construct a sectoral Climate Valuation Response (CVR), defined as the cumulative price-dividend impulse response over a 24-month horizon. This measure captures the long-run discount-rate adjustment induced by a climate uncertainty shock and provides a natural metric for comparing sectoral reactions as a reflection of their heterogeneous exposures to transition risk.[28]

[28] For instance, the Campbell–Shiller's approximate identity (Campbell and Shiller, 1988) implies that innovations in the log price–dividend ratio $pd_t \equiv p_t - d_t$ can be decomposed into revisions of expected future dividend growth and revisions in expected future returns,

$$\Delta pd_{t+1} \approx \text{cash flow news}_{t+1} - \text{discount rate news}_{t+1},$$

where discount-rate news is the revision in future expected returns induced by a shock. The literature has documented that the price-dividend ratio predicts returns over horizons ranging from months to a few years much more

To organise our cross-sectional analysis, we group sectors by their emission intensity score. For each $k = 1, \ldots, 5$, we compute the average CVR across the k most carbon-intensive industries (*Top-k*), and the k least emission-intensive industries (*Bottom-k*):[29]

$$CVR_{(k)} = \frac{1}{k} \sum_{i \in \text{group}(k)} CVR_i.$$

Two patterns clearly emerge from Table 7. First, the average CVR is negative for all Top-k groups, indicating that carbon-intensive sectors experience persistent valuation declines following a futures-implied climate shock, consistent with heightened transition-risk exposure and sustained increases in discount rates. Second, the Bottom-k groups exhibit strictly positive CVR values for all $k \leq 5$, pointing to a medium-run re-pricing of low-emission, transition-resilient industries. This contrast reflects the fact that climate uncertainty shocks shift investor demand toward sectors with lower regulatory and technological exposure to the transition, generating a *green re-pricing effect*.

k	**Top-k Sectors (sector added)**	**Emission Intensity**	**Average CVR (Top-k)**
1	Miscellaneous (Other)	0.672	−0.695
2	Utilities	0.371	−0.460
3	Food	0.359	−0.212
4	Steel Works	0.325	−0.127
5	Chemicals	0.324	−0.186
k	**Bottom-k Sectors (sector added)**	**Emission Intensity**	**Average CVR (Bottom-k)**
1	Chips	0.047	0.587
2	Banks	0.066	0.326
3	Toys	0.079	0.387
4	Insurance	0.085	0.233
5	Apparel	0.102	0.373

Table 7: Climate Valuation Responses Averaged by Emission Scores. Top-k vs Bottom-k Sectors. For each k, the table reports the average 24-month cumulative price-dividend response (CVR) to a climate-policy uncertainty shock across the k highest-emission sectors (Top-k) and across the k lowest-emission sectors (Bottom-k). Emission intensities are scaled by 1,000.

To provide a clear visual representation of the emission-based differential in climate valuation responses, Figure 16 plots the average CVR for the Top-k and Bottom-k emission groups, constructed cumulatively for $k = 1, \ldots, 5$. The Top-k curve aggregates the k most carbon-intensive sectors, while the Bottom-k curve aggregates the k least emission-intensive sectors, so that increasing the values for k expands the corresponding cluster. The figure also reports the spread (dashed red line), defined as the difference $CVR_{\text{Bottom-}k} - CVR_{\text{Top-}k}$. The pattern is highly asymmetric: the Top-k CVR remains persistently negative for all values of k, whereas the Bottom-k CVR remains

reliably than it predicts dividend growth; variance decomposition exercises—based on monthly vector autoregressions—quantify the relative contributions of discount-rate and cash-flow news to unexpected returns and valuation movements and find discount-rate news explains a large share of return innovations. This empirical evidence implies that valuation-ratio fluctuations are primarily driven by discount-rate news rather than cash-flow news, see Cochrane (2011) and references therein. In this section, we are therefore measuring the differential of the discount rate news term under the assumption that the climate uncertainty shocks do not represent cash flow news or that the green-brown differential in the cash flow news they induced is negligible.

[29] For expositional purposes we present results for values of k up to 5, but the emission-based gradient is fully robust to alternative choices of k: Bottom$k$ minus Top-k groupings deliver the same qualitative separation across all specifications.

positive for all values of k, resulting in a consistently positive spread. The magnitude of the spread, which ranges between approximately 0.36 and 1.28, indicates that carbon-intensive industries systematically load more heavily on the discount rate risk associated to climate uncertainty shocks. This provides direct support to a *green re-pricing mechanism*, whereby climate uncertainty induces stronger valuation adjustments for industries that are more exposed to investor reallocation flows away from the effects of transition risks.

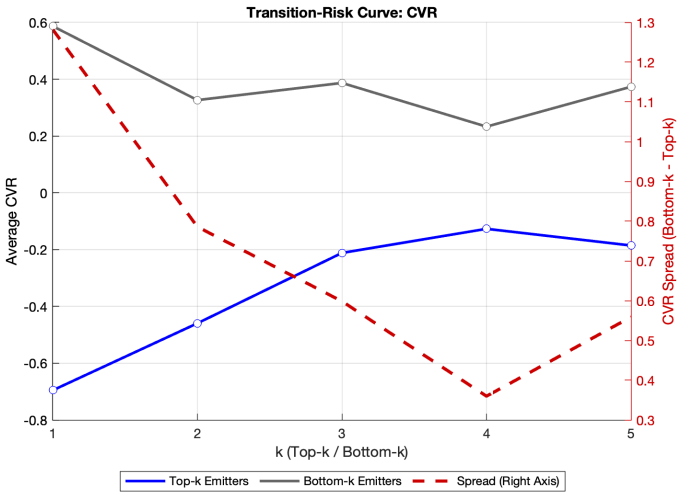


Figure 16
CVR curves. Transition-risk cluster CVR curves for the Top-k highest-emission sectors (blue) and the Bottom-k lowest-emission sectors (grey), plotted along with the corresponding spread (red, right axis).

To draw inference on the green–brown spread we estimate a VAR that includes the equal-weighted price–dividend ratios of the five least carbon-intensive sectors (Chips, Banks, Toys, Insurance, Apparel) and of the five most carbon-intensive sectors (Miscellaneous, Utilities, Food, Steel Works, Chemicals), together with VIX, IP, PCE, PD and FFR. We compute the difference between the two responses for each posterior draw (Figure 17).

The green–brown valuation spread is positive on impact and widens gradually, peaking at about 5 percentage points after eight months. Valuations of low-emission sectors therefore recover faster, while those of carbon-intensive sectors take about a year to return to their pre-shock level. If the shock carries similar cash-flow news for the two groups, this spread is consistent with climate uncertainty raising the discount rates applied to carbon-intensive activities by more than those applied to transition-resilient ones.

Together, the CVR asymmetry and the green-brown IRF spread reveal a consistent green re-pricing effect: climate uncertainty shocks differentially increase the discount rates applied to carbon-intensive activities and this shifts capital and investment towards transition-resilient industries. This re-pricing mechanism reinforces the segmentation of asset valuations along the emission spectrum over the medium run.

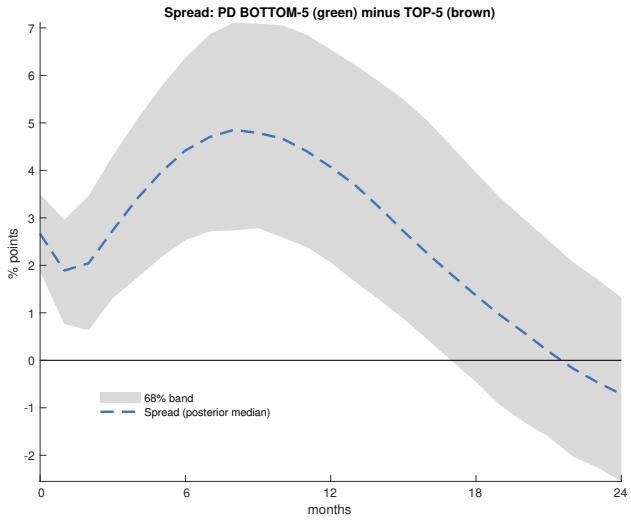


Figure 17
Green–brown PD spread. Difference between the responses of the equal-weighted price–dividend ratios of the five least and the five most carbon-intensive sectors, estimated in a single Bayesian Proxy-SVAR that includes both indices alongside VIX, IP, PCE, PD and FFR. The spread is computed draw by draw. Dashed line: posterior median. Shaded areas: 68% posterior credible intervals. IRFs in percentage points. Sample: February 2014–January 2025.

7 Climate Uncertainty Beyond Financial Volatility

The robustness checks of Section 5 show that our results are not sensitive to the construction of the instrument, the specification of the VAR, or individual episodes in the sample. A remaining concern is that the shock we identify captures general financial uncertainty rather than climate-policy uncertainty in particular. We address this concern directly by comparing the carbon-futures shock with a generic volatility shock, identified recursively by ordering the VIX first in the same BVARs. Both shocks are scaled to raise the VIX by one standard deviation on impact. Since both identifications are applied to the same posterior draws, the difference between the two sets of impulse responses can be computed draw by draw.

At the aggregate level the two shocks are not the same shock (Figure 18). With the same impact on the VIX, the generic shock fades more quickly, whereas the carbon-futures shock keeps volatility elevated for longer and is followed by a deeper fall in output and a larger reduction in the federal funds rate. The two shocks differ most in the response of prices. Under the generic shock the price level falls and stays below baseline throughout, the textbook response to an uncertainty shock; under the carbon-futures shock it falls by the same amount in the short run but rises above baseline after ten months. The medium-run increase in prices documented in Section 4 is therefore not shared by a generic volatility shock. One interpretation is that, once the precautionary fall in demand fades, the expected cost of a more uncertain regulatory path passes through to prices, as it does after carbon-price shocks (Känzig, 2023; Hensel et al., 2024). The aggregate valuations respond similarly to the two shocks, as one would expect of a variable that prices volatility itself.

The comparison is more informative at the sectoral level (Figures 19 and 20). The main difference concerns green sectors. After the carbon-futures shock their valuations rise above the pre-shock level by more than 4 percent, whereas after a generic volatility shock they remain below it over the entire horizon. As a result, under the generic shock the green–brown valuation spread becomes positive only after a few months and is short-lived; over the first year it is less than a

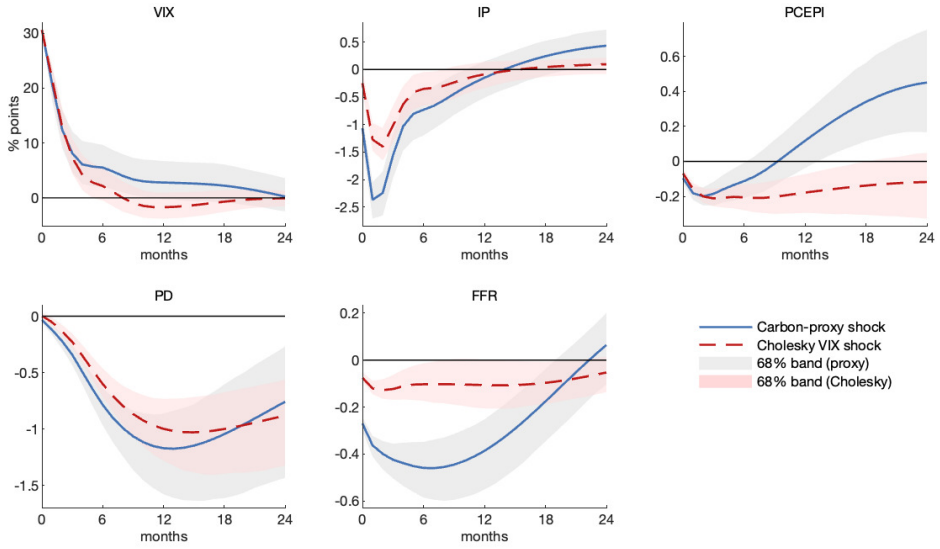


Figure 18
Aggregate Responses: Carbon-Futures Shock vs. a Generic Volatility Shock. Impulse responses to the carbon-futures shock (solid, 68% bands) and to a volatility shock identified recursively with the VIX ordered first (dashed, 68% bands), both normalised to a 30.65 percent impact increase of the VIX. IRFs in percentage points.

third of the spread generated by the carbon-futures shock. The green re-pricing documented in Section 6 is a feature of climate-policy uncertainty, not of financial volatility. Output tells the same story: green industries respond identically to the two shocks, and the entire additional contraction induced by the carbon-futures shock falls on brown industries. Panel (c) of each figure plots the difference between the two estimated green–brown spreads, computed draw by draw, together with its credible intervals. The difference is positive and significant over most of the first year and a half, for both valuations and output. A volatility shock of the same size therefore accounts for only part of the sectoral response; the remainder is specific to the carbon-futures shock.

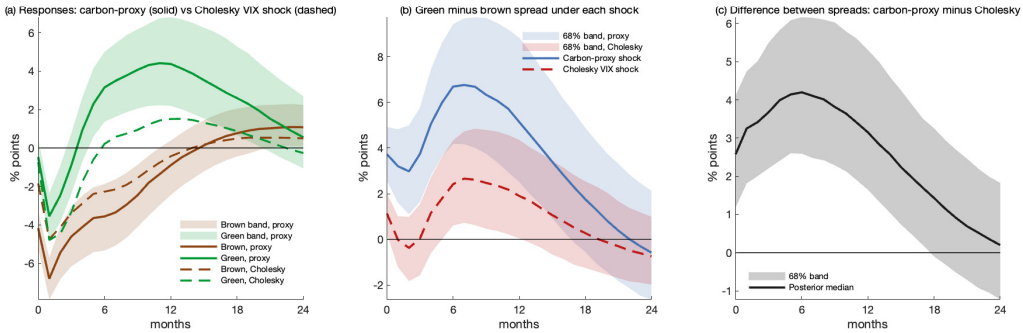


Figure 19
Sectoral Valuations: Carbon-Futures Shock vs. a Generic Volatility Shock. (a) Brown and green price–dividend responses to the carbon-futures shock (solid, 68% bands) and to a recursive VIX shock of the same impact size (dashed). (b) Green minus brown spread under each shock. (c) Difference between the two spreads. Draw-by-draw computations on the same posterior draws.

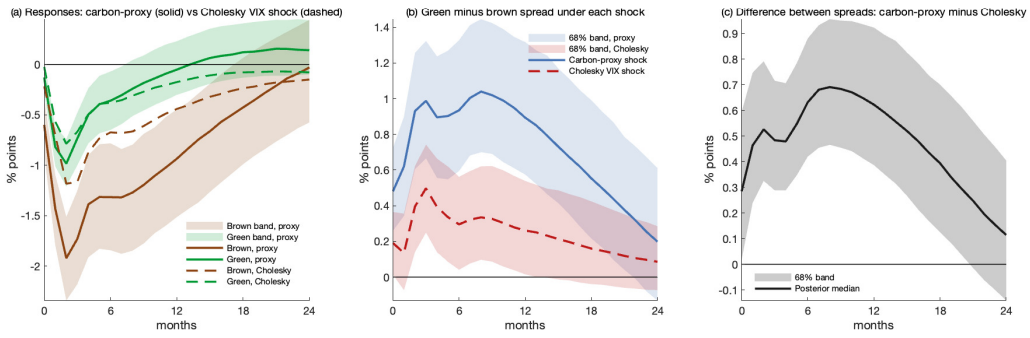


Figure 20
Sectoral Output: Carbon-Futures Shock vs. a Generic Volatility Shock. (a) Brown and green industrial production responses to the carbon-futures shock (solid, 68% bands) and to a recursive VIX shock of the same impact size (dashed). (b) Green minus brown industrial productions under each shock. (c) Difference between the two spreads. Draw-by-draw computations on the same posterior draws.

8 Conclusion

This paper provides new evidence on the macro-financial effects of climate-related uncertainty in the United States. Using a monthly Bayesian Proxy-SVAR identified with an external instrument derived from the realized volatility of ICE U.S. Carbon Futures log-price changes, we document how unexpected increases in transition risk tighten financial conditions and propagate to the real economy.

A rise in climate uncertainty leads to a marked increase in the VIX, accompanied by declines in aggregate industrial production and equity market valuations, as well as a loosening of monetary policy through lower policy rates. Consumer prices decline in the short run, suggesting that demand effects dominate in the months following the shock, and rise above their pre-shock level over the medium run. The initial deflationary response indicates that our proxy mainly captures the *second-moment dimension of uncertainty* rather than direct policy actions on carbon pricing, whilst the subsequent increase in the price level goes in the direction of the first-moment effects reported in the literature. Whereas actual climate-policy interventions that raise carbon prices tend to be inflationary—reflecting higher production costs, tighter energy supply, and cost-push pressures—our measure captures instead the uncertainty surrounding such actions, which primarily affects expectations and financial conditions in the short run.

Notably, the adjustment dynamics of financial variables, such as the VIX, the equity price-dividend ratio, and the policy short-term rate, appears considerably more persistent than that of the macroeconomic variables like industrial production and consumer prices. This persistence is consistent with a financial transmission channel through which climate uncertainty spreads across the economy, operating via risk re-pricing, shifts in asset valuations, and changing financial conditions. Overall, these dynamics suggest that climate policy uncertainty acts as a financial amplifier of macroeconomic fluctuations. These results remain robust across a range of specifications, including alternative lag structures and prior choices, and persist when the instrument is purged of carbon-price, energy-market, policy-uncertainty and financial-market components.

Next, we classify sectors according to their exposure to transition risk: *brown* industries include energy-supply and fossil-fuel-based activities such as utilities, oil and gas extraction, and petroleum refining; *green* sectors comprise low-emission and transition-enabling manufacturing, such as elec-

trical equipment and electronic components. At the sectoral level, we uncover a pronounced *green–brown asymmetry* in the transmission of climate uncertainty shocks. Carbon-intensive industries experience deeper and more persistent contractions in output, while low-emission and technology-oriented sectors display greater resilience and faster recovery. In financial markets, equity valuations decline on impact for all industries but rebound and overshoot in the case of green sectors, suggesting a reallocation of capital toward transition-resilient assets. These findings support the view that the asymmetric responses of the two groups are particularly important, as they highlight how transition uncertainty shocks redistribute risks and investment opportunities across the economy, by accelerating capital flows toward greener industries while exposing carbon-intensive activities to greater financial vulnerability.

To complement this broad two-sector decomposition, we also estimate a fully granular, sector-by-sector specification. The granular sector-by-sector analysis further strengthens these conclusions. By linking each industry to its EPA supply-chain emission intensity, we uncover a clear cross-sectional gradient: sectors characterized by higher carbon intensity exhibit systematically larger and more persistent real and financial contractions following a climate uncertainty shock. This confirms and extends the green–brown asymmetry documented earlier, showing that the severity of climate-induced adjustments increases monotonically with emission intensity.

Methodologically, our contribution lies in developing and exploiting a market-based proxy of climate uncertainty derived from the estimated *innovation* of an autoregressive process applied to realized carbon futures returns realized variance. This innovation-based proxy captures unexpected revisions in investors' expectations about the credibility and timing of climate regulations, offering a cleaner identification of the transition uncertainty channel. By linking carbon market dynamics to broader macro-financial conditions, our approach bridges the literatures on climate-policy risk, financial transmission, and macroeconomic volatility.

In a policy perspective, our findings emphasize the importance of predictable and credible climate policies in mitigating the macro-financial costs of the energy transition. Our results indicate that uncertainty about the timing and stringency of future regulation operates as an amplifier of macroeconomic fluctuations, raising risk premia, depressing investment, and disproportionately affecting carbon-intensive industries. This aligns with theoretical analyses showing that regulatory ambiguity can delay or distort firms' adjustment decisions, leading to inefficient capital stranding, higher transition costs, and lower welfare (Lemoine and Traeger (2016); Barnett et al. (2020)). In financial markets, the heightened volatility and risk re-pricing associated with transition uncertainty echo the asset-pricing evidence that carbon-intensive firms command higher required returns due to their exposure to downside policy risk (Bolton and Kacperczyk (2021); Bolton and Kacperczyk (2023)), while predictable regulatory trajectories help compress risk premia and support the financing of green investment.[30] Together with our empirical findings, this body of work suggests that by anchoring expectations and alleviating the second-moment shocks that propagate through financial markets, credible climate policy can therefore sustain investment, reduce transition risk premia, and promote a more orderly, efficient, and financially stable path to decarbonisation.

[30] At the macro level, general equilibrium models indicate that policy credibility is a key determinant of the pace and cost of decarbonisation. When agents have confidence in a clear long-run policy path, the transition unfolds with smoother capital reallocation, lower precautionary savings, and more sustained innovation (see Acemoglu et al. (2016); Van der Ploeg and Rezai (2020)).

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Online Appendix A: Sources of the Data

1 Data Source

Country: United States Monthly Frequency — February 2014 to January 2025		
Variable	**Definition**	**Construction and Source**
$\mathbf{VIX}_t$	CBOE Volatility Index. Benchmark measure of aggregate financial uncertainty implied by S&P 500 options.	Monthly average of daily VIX values. Measures expected equity market volatility over a 30-day horizon. Source: Federal Reserve Economic Data (FRED).
$\mathbf{RVUS}_t$	Realized volatility of the ICE U.S. Carbon Futures Index, used as a market-based proxy for climate policy uncertainty. In the baseline specification we employ the U.S. carbon futures index, which aggregates price dynamics from both the Western Climate Initiative (WCI) and the Regional Greenhouse Gas Initiative (RGGI).	Computed from daily log-returns of ICE U.S. Carbon Futures prices. Monthly realized volatility is defined as: $$RV_m = \sqrt{\frac{30}{n_m}\sum_{d=1}^{n_m} r_d^2}$$ where n_m denotes the number of trading days in month m. Price data are obtained from Bloomberg. Futures contracts reference emission allowances from both the WCI and RGGI markets. As a robustness exercise, we alternatively use the WCI-specific futures index, which exhibits a statistically significant first-stage F-statistic when employed as an instrument.
ε_t^{RV}	Climate uncertainty shock (external instrument).	Innovation component of carbon futures realized volatility, constructed as the residual from: $$RV_t = \alpha + \beta RV_{t-1} + \varepsilon_t$$ Estimated over the full sample. The residual isolates unanticipated, forward-looking revisions in climate-policy uncertainty.
$\mathbf{IP}_t$	Industrial Production Index. Measure of real economic activity.	Seasonally adjusted aggregate industrial production index from the Federal Reserve G.17 release. Source: Federal Reserve Economic Data (FRED).
$\mathbf{UR}_t$	Unemployment Rate. Measure of labor market slack.	US Unemployment rate, seasonally adjusted. Percent. Monthly Frequency. Source: Federal Reserve Economic Data (FRED).

Table (continued)

Variable	Definition	Construction and Source
$\mathbf{IP}_{s,t}$	Sectoral Industrial Production (NAICS-based).	Sector-level IP series from the Federal Reserve G.17 dataset. Industries classified according to NAICS codes. Green and brown aggregates constructed as equal-weighted averages of selected sectors.
$\mathbf{PCE}_t$	Personal Consumption Expenditures Price Index. Inflation measure.	Chain-type PCE price index, monthly and seasonally adjusted. Source: Federal Reserve Economic Data (FRED).
$\mathbf{FFR}_t$	Effective Federal Funds Rate. Monetary policy instrument.	Percent. Monthly average of the effective federal funds rate. Source: Federal Reserve Economic Data (FRED).
$\mathbf{PD}_t$	Aggregate S&P 500 price–dividend ratio. Forward-looking valuation measure.	Computed from Robert Shiller's S&P 500 prices and dividends. Dividends computed over a trailing 12-month window. Only portfolios with at least ten firms in every month of the sample are used. Source: Shiller Dataset. https://shillerdata.com
$\mathbf{SP500}_t$	S&P 500 stock market index. Broad measure of U.S. equity prices.	Monthly S&P 500 index level from Robert Shiller's dataset. Used as an alternative valuation indicator to the price–dividend ratio in robustness checks. Source: Shiller Dataset. https://shillerdata.com
$\mathbf{EPU}_t$	Economic Policy Uncertainty Index. Broad measure of policy-related uncertainty.	News-based index constructed by Baker et al. (2016), capturing the frequency of newspaper articles referring to economic policy uncertainty. Monthly frequency. Source: policyuncertainty.com.
$\mathbf{WTI}_t$	West Texas Intermediate crude oil price. Proxy for energy market conditions.	Logarithm of the spot price of WTI crude oil, measured in U.S. dollars per barrel. Monthly frequency. Source: Federal Reserve Economic Data (FRED).
$\mathbf{PD}_{s,t}$	Sectoral price–dividend ratios (SIC-based).	Constructed from Fama–French 49 Industry Portfolios. Dividends computed over a trailing 12-month window. Source: Kenneth French Data Library.

Table (continued)

Variable	Definition	Construction and Source
Emissions$_s$	Sectoral supply-chain greenhouse gas emission intensity.	EPA supply-chain emission factors by NAICS (kg CO_2e per USD of output). Sector-level intensities obtained as averages of corresponding 6-digit NAICS codes. Source: U.S. Environmental Protection Agency. U.S. Environmental Protection Agency.

Data sources, definitions, and construction of variables. This table reports the variables used in the analysis, together with their definitions, construction details, and original data sources.

Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max	N
VIX_t	2.8342	0.3065	2.3155	4.0560	132
IP_t	4.6141	0.0304	4.4389	4.6454	132
PCE_t	4.6671	0.0843	4.5711	4.8317	132
PD_t	3.9877	0.2019	3.5784	4.3211	132
FFR_t	0.0166	0.0186	0.0005	0.0533	132
$SP500_t$	8.0000	0.3600	7.5050	8.7013	132
UR_t	0.0528	0.0205	0.0340	0.1480	132
EPU_t	4.9800	0.3900	4.1570	6.2225	132
WTI_t	4.0500	0.3300	2.8317	4.7421	132
$RVUS_t$	0.0927	0.0456	0.0210	0.3276	132
$RV_{WCI,t}$	0.0405	0.0305	0.0016	0.2805	132
ε_t^{RV}	0.0005	0.0285	-0.0644	0.2104	132
$\varepsilon_t^{RV,AR(2)}$	0.0006	0.0288	-0.0750	0.2110	132
$\varepsilon_t^{RV,ARMA(1,1)}$	0.0010	0.0295	-0.0788	0.2124	132
$\varepsilon_t^{RV,WCI}$	0.0010	0.0300	-0.0841	0.2573	132

Table 9: Summary Statistics.

The table reports descriptive statistics for the main variables employed in the empirical analysis. The first block refers to the variables included in the baseline (aggregate) VAR specification. The second block reports alternative non-sectoral specifications used in robustness exercises, including the S&P 500 index, the unemployment rate, the Economic Policy Uncertainty (EPU) index of Baker et al. (2016), and the log WTI crude oil price. The third block reports the level of realized volatility for the U.S. carbon futures index (RVUS) and for the Western Climate Initiative (WCI) carbon market. The final block reports the external instrument, defined as the innovation in carbon futures realized volatility, along with alternative specifications based on AR(2), ARMA(1,1), and WCI-based measures. All variables enter the model in natural logarithms, except for the federal funds rate (FFR) and the unemployment rate (UR), which are expressed in levels (percent). PD denotes the log price–dividend ratio, while SP500 corresponds to the log level of the S&P 500 index.

Online Appendix B: Robustness to the Number of Lags

This appendix assesses the robustness of the impulse response functions to alternative choices of the BVAR lag length. In addition to the baseline specification with six lags, we re-estimate the

model using three, twelve, and twenty-four lags. The comparison allows us to evaluate whether the dynamic responses to the identified shock are sensitive to the lag structure of the VAR.

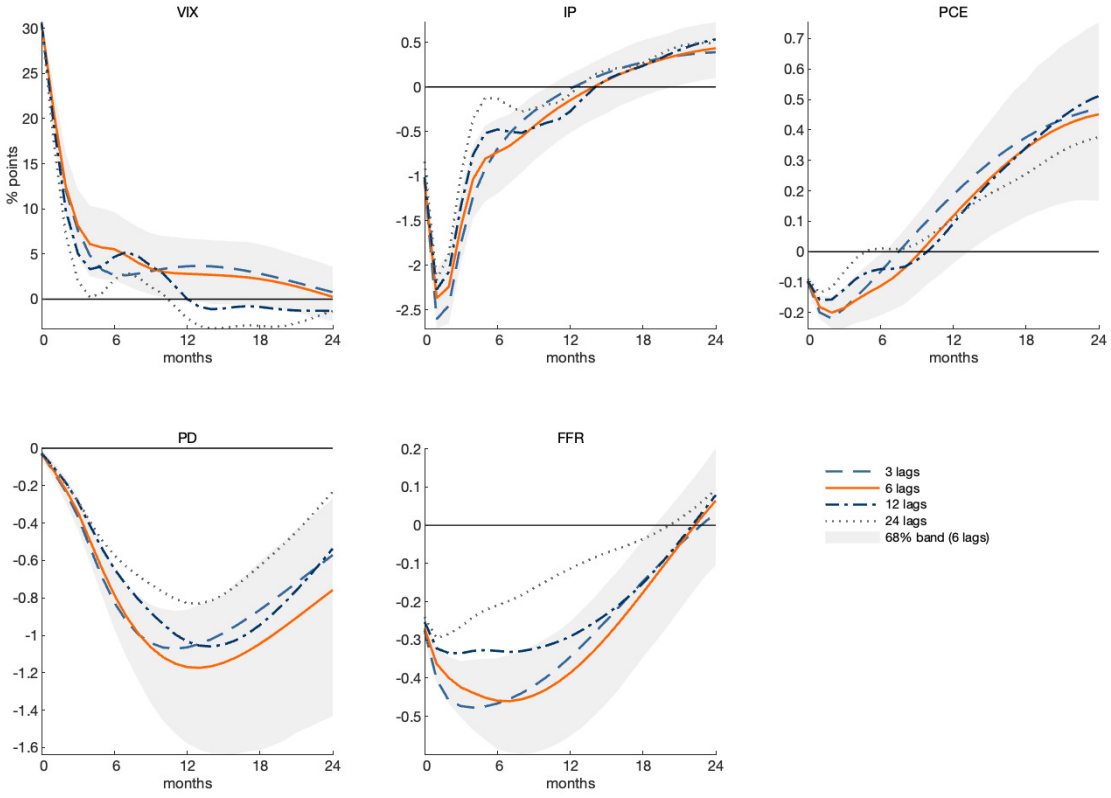


Figure 21
Robustness to the number of Lags. Impulse response functions estimated using alternative VAR lag lengths (3, 6, 12, and 24 lags). Solid and dashed lines report point estimates for each specification. Shaded areas denote 68% posterior credible intervals for the baseline specification with six lags. IRFs are reported in percentage points. Sample: February 2014–January 2025.

Overall, the impulse responses are highly robust across alternative lag specifications. The magnitude, sign, and timing of the responses are remarkably similar for the 3-, 6-, and 12-lag models across all variables. When using 24 lags, some attenuation in the responses emerges, most notably for the federal funds rate, whose path lies partially outside the 68% credible interval of the baseline specification at intermediate horizons. This behavior is consistent with the loss of degrees of freedom and weaker parameter precision associated with highly parameterized monthly VARs. Importantly, even in this case, the qualitative dynamics—including the direction of the responses and their overall persistence—remain unchanged. Taken together, these results indicate that the baseline findings are not driven by the choice of lag length and are robust to alternative dynamic specifications.

Online Appendix C: Alternative Market-Based Surprise Measures

WCI Surprises

We assess the robustness of our main findings by replacing the baseline realized-volatility innovation with a WCI-based measure. The identification strategy remains unchanged. We extract the

unexpected component from the AR(1) process:

$$RV_t^{WCI} = \alpha + \beta RV_{t-1}^{WCI} + \varepsilon_t,$$

where ε_t denotes the innovation. The residuals are used as the external instrument.

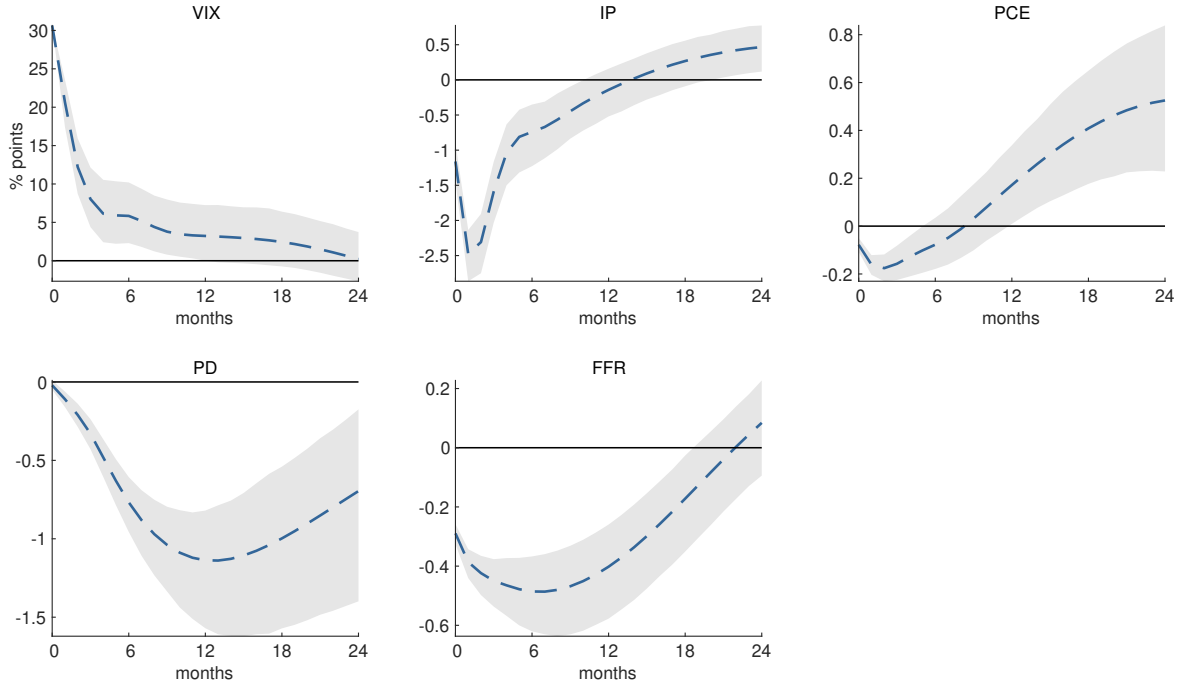


Figure 22
WCI Surprises. Shaded areas denote 68% posterior credible intervals. IRFs are reported in percentage points. Sample: February 2014–January 2025.

The resulting impulse responses are quantitatively and qualitatively similar to the baseline.

AR(2)-Based ICE Surprise

We further allow for richer dynamics in the construction of the ICE surprise. Instead of the baseline AR(1),

$$RV_t = \alpha + \beta RV_{t-1} + \varepsilon_t,$$

we estimate:

$$RV_t = \alpha + \beta_1 RV_{t-1} + \beta_2 RV_{t-2} + \varepsilon_t.$$

The residuals ε_t are used as the external instrument. The first-stage F-statistic equals 16.66, indicating instrument relevance.

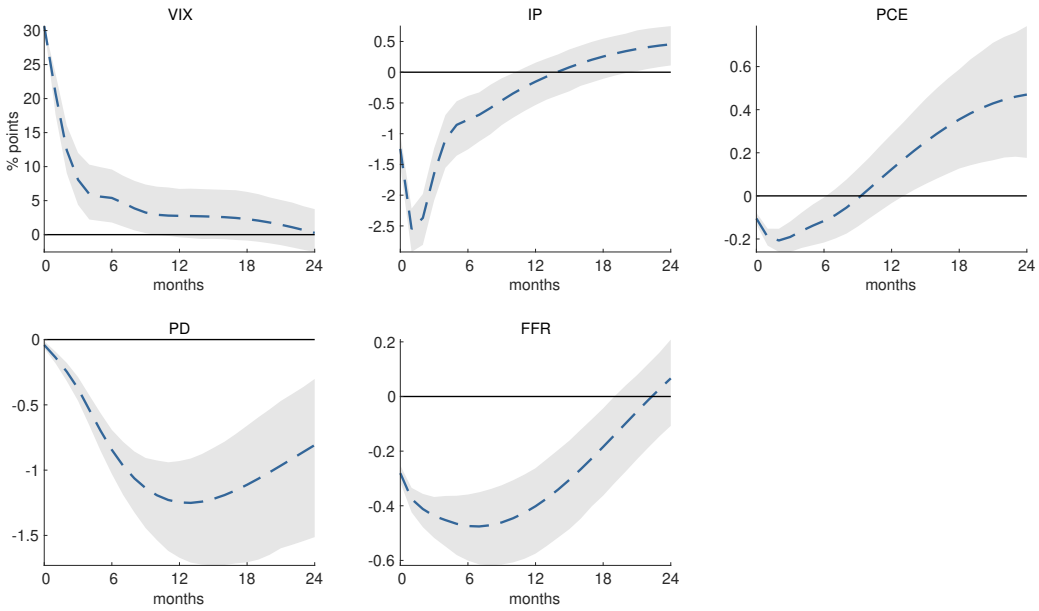


Figure 23
AR(2)-Based ICE Surprises. Shaded areas denote 68% posterior credible intervals. IRFs are reported in percentage points. Sample: February 2014–January 2025.

Results are unchanged relative to the baseline specification.

ARMA(1,1)-Based ICE Surprise

Alternatively, we construct the surprise measure allowing for both autoregressive and moving-average dynamics. Specifically, we estimate the following ARMA(1,1) process:

$$RV_t = \alpha + \beta RV_{t-1} + \theta \varepsilon_{t-1} + \varepsilon_t,$$

where ε_t is the innovation. The residuals ε_t are used as the external instrument in the Bayesian Proxy-SVAR framework. The first-stage F-stat equals 16.15, confirming instrument relevance.

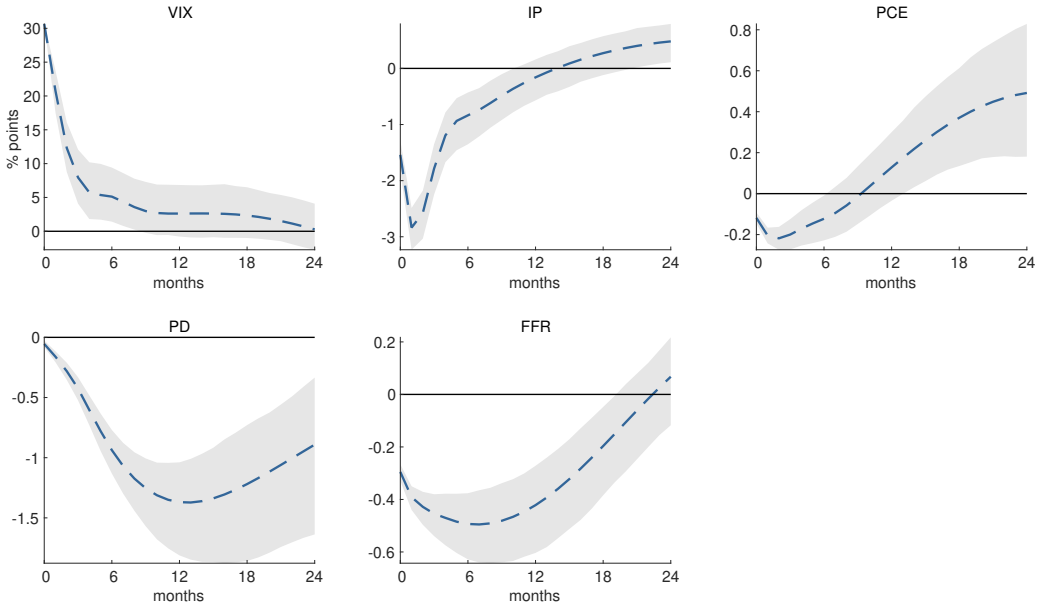


Figure 24
ARMA(1,1)-Based ICE Surprises. Shaded areas denote 68% posterior credible intervals. IRFs are reported in percentage points. Sample: February 2014–January 2025.

Impulse responses are virtually unchanged relative to the baseline specification.

Appendix D: Climate-Policy Event List

Table 10: Climate-policy events. Date on which the information became public and primary source. The table lists the 79 events other than allowance auctions. The remaining 88 events are the result releases of all quarterly RGGI and California–Québec allowance auctions held between 2014 and 2024, dated from RGGI, Inc. press releases and CARB auction reports.

Date	Event	Source
Decisions on the two regional programs		
2014-04-25	CARB Board approves comprehensive cap-and-trade amendments	CARB
2014-09-18	CARB Board approves cap-and-trade amendments at public hearing	CARB
2016-09-08	Governor Brown signs SB 32	Gov. of California
2017-04-06	Cal. Chamber of Commerce v. CARB: Court of Appeal upholds cap-and-trade auctions	Cal. Courts
2017-06-28	Cal. Chamber of Commerce v. CARB: California Supreme Court denies review	Cal. Supreme Court docket
2017-07-17	AB 398 (cap-and-trade extension to 2030) passes Legislature	Cal. Legislature
2017-07-25	Governor Brown signs AB 398	Gov. of California
2017-07-27	CARB Board approves cap-and-trade amendments	CARB
2017-08-23	RGGI states announce proposed 2017 Program Review changes	RGGI, Inc.
2017-09-22	California, Quebec and Ontario sign linkage agreement	Gov. of California
2017-11-09	Virginia DEQ proposed CO2 trading rule linking to RGGI	RGGI, Inc.
2017-12-14	CARB Board approves 2017 Climate Change Scoping Plan	CARB
2017-12-19	RGGI states release updated 2017 Model Rule	RGGI, Inc.
2018-01-29	New Jersey Gov. Murphy signs Executive Order 7 to rejoin RGGI	State of New Jersey
2018-06-15	Ontario Premier-designate Ford announces cancellation of cap-and-trade	CARB
2018-07-03	Ontario revokes cap-and-trade regulation O.Reg.144/16	Gov. of Ontario
2018-12-13	CARB Board adopts cap-and-trade amendments implementing AB 398	CARB
2018-12-17	New Jersey proposes RGGI re-entry rule	RGGI, Inc.
2019-04-19	Virginia Air Board approves final CO2 trading regulation	RGGI, Inc.
2019-05-06	Virginia budget language blocks RGGI participation	RGGI, Inc.
2019-06-17	New Jersey adopts rules to rejoin RGGI	RGGI, Inc.
2019-10-03	Pennsylvania Gov. Wolf Executive Order 2019-07 directing DEP to join RGGI	Commonwealth of Pa.
2019-10-23	United States v. California: DOJ lawsuit against California-Quebec linkage filed	DOJ
2020-03-12	United States v. California: court rejects Treaty Clause and Compact Clause claims	Court filing (GovInfo)
2020-07-08	Virginia finalizes CO2 Budget Trading regulation; joins RGGI from 2021	RGGI, Inc.
2020-07-17	United States v. California: summary judgment for California on remaining claim	Court filing (GovInfo)
2021-03-15	RGGI announces Third Adjustment for Banked Allowances	RGGI, Inc.
2021-07-13	Pennsylvania Environmental Quality Board adopts final RGGI rule	Pa. DEP
2021-09-01	Pennsylvania IRRC approves RGGI rule	Pa. IRRC
2022-01-15	Virginia Gov. Youngkin executive order (EO 9) to begin withdrawing Virginia from RGGI	Gov. of Virginia
2022-04-23	Pennsylvania RGGI rule published in Pa. Bulletin	Pa. Bulletin
2022-07-08	Pa. Commonwealth Court preliminarily enjoins Pennsylvania RGGI rule	Pa. Courts
2022-12-15	CARB Board approves 2022 Scoping Plan	CARB
2023-06-07	Virginia Air Pollution Control Board votes to repeal RGGI regulation	Gov. of Virginia
2023-11-01	Pa. Commonwealth Court (Bowfin KeyCon v. DEP) rules RGGI rule an unconstitutional tax	Pa. Courts
2023-11-21	Shapiro administration appeals Bowfin ruling to Pa. Supreme Court	Court document
2024-07-10	CARB workshop: cap-and-trade amendments delayed	CARB (workshop notice)
2024-09-23	RGGI Third Program Review update	RGGI, Inc.
2024-11-20	Floyd County (VA) Circuit Court rules Virginia RGGI repeal unlawful	Court document
Federal and international policy		
2014-06-02	EPA proposes Clean Power Plan	Federal Register

(continued)

Date	Event	Source
2015-08-03	Clean Power Plan final rule announced	Federal Register
2015-12-12	Paris Agreement adopted at COP21	UNFCCC
2016-02-09	Supreme Court stays Clean Power Plan	U.S. Supreme Court
2016-04-22	United States signs Paris Agreement	UNFCCC
2016-09-03	United States joins Paris Agreement	White House (archive)
2017-10-10	EPA proposes repeal of Clean Power Plan	Federal Register
2018-08-21	EPA proposes Affordable Clean Energy (ACE) rule	Federal Register
2019-06-19	EPA finalizes ACE rule and repeal of Clean Power Plan	EPA
2020-03-31	EPA/NHTSA announce final SAFE Vehicles Rule	EPA
2020-11-04	US withdrawal from Paris Agreement takes effect	State Dept. (archive)
2021-01-19	D.C. Circuit vacates ACE rule	Court filing (GovInfo)
2021-02-19	US officially rejoins Paris Agreement	State Dept. (archive)
2021-09-20	Climate Week NYC 2021 opens	Climate Week NYC
2021-11-19	House passes Build Back Better Act	Congress.gov
2021-12-19	Sen. Manchin says he cannot vote for Build Back Better	U.S. Senate
2022-06-30	Supreme Court decides West Virginia v. EPA	U.S. Supreme Court
2022-07-14	Manchin tells Schumer he will not support climate spending in reconciliation	Press report
2022-07-27	Schumer-Manchin announce Inflation Reduction Act deal	U.S. Senate
2022-08-07	Senate passes Inflation Reduction Act	Congress.gov
2022-08-12	House passes Inflation Reduction Act	Congress.gov
2022-08-16	President Biden signs Inflation Reduction Act	Congress.gov
2023-05-11	EPA proposes GHG standards for fossil-fuel power plants	EPA
2024-04-25	EPA finalizes carbon pollution standards for power plants	EPA
2024-11-24	COP29 concludes with climate-finance deal	UNFCCC
Presidential elections and executive actions		
2016-11-08	US presidential election 2016	FEC
2017-01-20	Trump inauguration; America First Energy Plan replaces Climate Action Plan	White House (archive)
2017-01-24	Trump memoranda advancing Keystone XL and Dakota Access pipelines	Federal Register
2017-03-28	Executive Order 13783 orders review of the Clean Power Plan	Federal Register
2017-06-01	President Trump announces US withdrawal from Paris Agreement	White House (archive)
2019-11-04	US submits formal notification of withdrawal from Paris Agreement	State Dept. (archive)
2020-11-03	US presidential election 2020	FEC
2021-01-20	President Biden signs instrument to rejoin Paris Agreement	White House (archive)
2021-01-27	Executive Order 14008 on the climate crisis	Federal Register
2021-04-22	Leaders Summit on Climate: new U.S. target of 50–52% below 2005 by 2030	White House (archive)
2024-11-05	US presidential election 2024	FEC
2025-01-20	Executive order withdrawing US from Paris Agreement	White House
Midterm elections		
2014-11-04	US midterm elections 2014	Clerk of the House
2018-11-06	US midterm elections 2018	Clerk of the House
2022-11-08	US midterm elections 2022	Clerk of the House